

Hellenic Complex Systems Laboratory

Uncertainty Estimation of Diagnostic Accuracy Measures under Parametric Distributions

Technical Report XXIX

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2025

Uncertainty Estimation of Diagnostic Accuracy Measures under Parametric Distributions

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DOI: [10.5281/zenodo.22963267](https://doi.org/10.5281/zenodo.22963267)

Abstract

Background: Diagnostic tests are evaluated by their ability to discriminate between diseased and nondiseased individuals; however, estimates of diagnostic accuracy alone do not quantify their associated uncertainty. Uncertainty assessment is therefore essential for interpreting accuracy estimates and comparing diagnostic thresholds.

Methods: We develop a parametric framework for propagating uncertainty to diagnostic accuracy measures formulated as functions of sensitivity, specificity, and prevalence. Normal, lognormal, or gamma models are specified separately for the diseased and nondiseased populations. At a diagnostic threshold, first-order uncertainty propagation retains covariance between estimated population means and standard deviations, incorporates prevalence sampling uncertainty when applicable, and adds the propagated measurement variance to the sampling variance while retaining measurement covariance between sensitivity and specificity. The Wolfram Language implementation provides uncertainty components by source, joint output covariance, and approximate expanded-uncertainty intervals calculated using effective degrees of freedom derived from the source-specific variance contributions. At estimated optimal thresholds, measurement variance is evaluated by Gauss–Hermite quadrature, conditional on the selected threshold; optimal threshold uncertainty is excluded.

Results: Illustrative analyses indicate that the contribution of each uncertainty source varies with the diagnostic accuracy measure, distributional assumptions, and threshold. Retaining measurement covariance can reduce the propagated measurement variance of a derived measure. In the cases examined, joint propagation reproduces the corresponding marginal variances and enables evaluation of uncertainty across diagnostic thresholds.

Conclusions: The framework provides a reproducible basis for assessing diagnostic accuracy and uncertainty while preserving relevant input and output dependencies. Decomposition by source helps identify the assumptions and uncertainty components underlying each estimate. The intervals remain local approximations; nominal coverage and optimal threshold uncertainty require separate assessment.

Keywords:

diagnostic accuracy; covariance propagation; measurement uncertainty; sampling uncertainty; diagnostic threshold; delta method; lognormal distribution; effective degrees of freedom; Wolfram Language

1. Introduction

Medical diagnosis integrates clinical evidence with a model of disease status [1]. Quantitative tests classify measurements relative to a diagnostic threshold. Their performance depends on the measurement distributions in diseased and nondiseased populations and on the population in which the test is applied [2–4].

Diseased and nondiseased populations are denoted by D and H. At the specified diagnostic threshold t , results are positive when $X \geq t$ and negative when $X < t$. Figure 1 illustrates overlapping population distributions; Table 1 defines the corresponding contingency table.

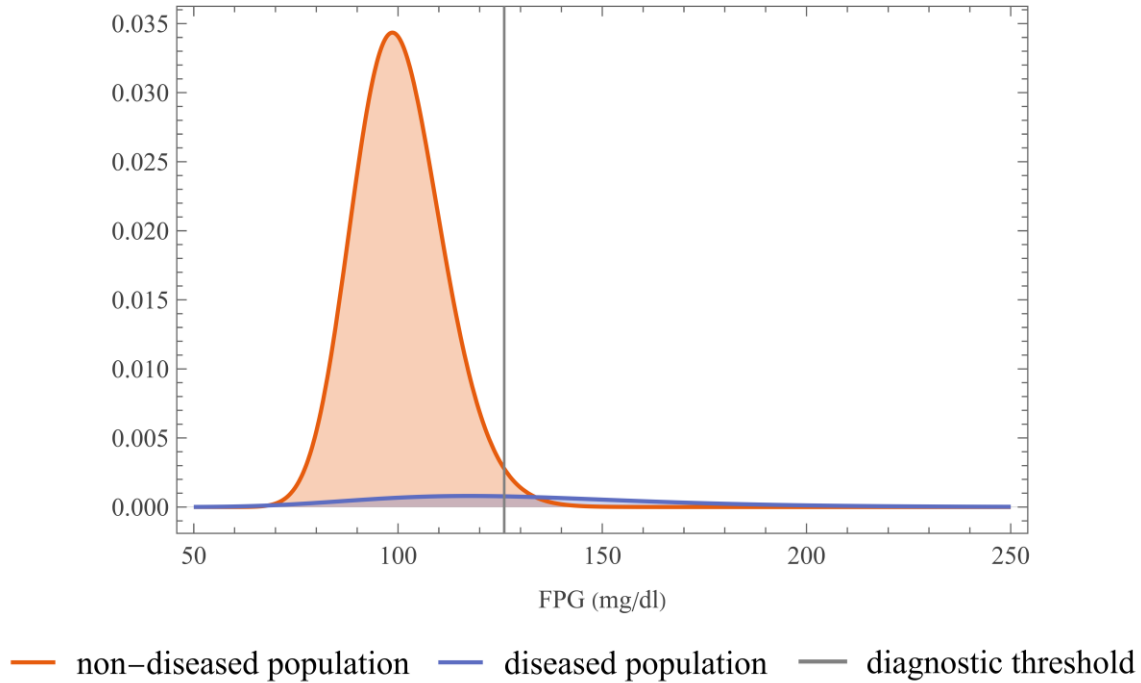


Figure 1. Lognormal probability densities for the supplied FPG inputs at an illustrative threshold of 126 mg/dL.

Table 1. Contingency table for a positive-above-threshold test.

Test result	Diseased (D)	Nondiseased (H)
Positive ($X \geq t$)	TP	FP
Negative ($X < t$)	FN	TN

1.1.Diagnostic accuracy measures

The sixteen measures are sensitivity (Se), specificity (Sp), overall diagnostic accuracy (ACC), positive and negative predictive values (PPV, NPV), diagnostic odds ratio (DOR), positive and negative likelihood ratios (LR^+ , LR^-), Youden’s index (J), balanced accuracy (BA), concordance probability (CZ), Fowlkes–Mallows index (FMI), Cohen’s kappa (κ), prevalence- and bias-adjusted kappa (PABAK), F_1 score (F_1), and Matthews correlation coefficient (MCC) [4, 5].

Table 2. Interpretation and admissible range of the sixteen diagnostic accuracy measures.

Measure	Interpretation	Range
Se	Proportion positive among diseased individuals.	$[0, 1]$
Sp	Proportion negative among nondiseased individuals.	$[0, 1]$
ACC	Prevalence-weighted probability of correct classification.	$[0, 1]$
PPV	Probability of disease conditional on a positive result.	$[0, 1]$
NPV	Probability of no disease conditional on a negative result.	$[0, 1]$
DOR	Ratio of the odds of a positive result in the two populations.	$[0, \infty)$
LR ⁺	Se divided by the false-positive probability.	$[0, \infty)$
LR ⁻	False-negative probability divided by Sp.	$[0, \infty)$
J	Se plus Sp minus one.	$[-1, 1]$
BA	Arithmetic mean of Se and Sp.	$[0, 1]$
CZ	Product of Se and Sp.	$[0, 1]$
FMI	Geometric mean of Se and PPV.	$[0, 1]$
κ	Agreement corrected for the chance agreement from both marginals.	$[-1, 1]$
PABAK	$2 \text{ ACC} - 1$.	$[-1, 1]$
F_1	Harmonic mean of Se and PPV.	$[0, 1]$
MCC	Correlation between binary test classification and disease status.	$[-1, 1]$

Increasing the threshold decreases Se and increases Sp under this decision rule. This opposing relationship between sensitivity and specificity does not alone determine a clinically optimal threshold. PPV, NPV, FMI, κ , PABAK, F_1 , and MCC also depend on prevalence. Operational decisions may depend on consequences and costs [3].

1.2. Uncertainty

Standard uncertainty expresses uncertainty on a standard-deviation scale. For a derived measure, specified input uncertainties and covariances are propagated through the measurement model [6, 7]. Uncertainty in diagnostic accuracy measures is therefore conditional on the model, sampling design, and represented uncertainty sources.

Total sampling variance is the sum of the population-parameter and prevalence sampling variance contributions. Under the stated independence assumptions, combined variance additionally includes the propagated measurement variance; combined standard uncertainty is the nonnegative square root of this sum.

Measurement uncertainty is propagated jointly through both distribution tails at the diagnostic threshold, with the fitted distributions held fixed. The measurement sensitivity coefficients for Se and Sp have opposite signs, yielding a nonpositive covariance contribution that must be retained when propagating uncertainty to measures that depend on both [6].

The diagnostic threshold t is a specified fixed decision value. For a diagnostic accuracy measure F , $u_c^2(F) = u_s^2(F) + u_m^2(F)$, where $u_c(F)$, $u_s(F)$, and $u_m(F)$ denote combined, total sampling, and propagated measurement standard uncertainty, respectively. This variance decomposition assumes that the sampling and measurement uncertainty sources are independent.

Treating the measurement contributions to Se and Sp as independent omits their measurement covariance. Such a model requires separate justification, uncertainty magnitudes, and dependence assumptions. Even with identical marginal uncertainties, it generally yields different uncertainties for measures combining the two rates. The present model retains this covariance, conditional on the specified population distributions.

Accounting for measurement uncertainty can affect the inferred population distributions, including their tails, and estimates of diagnostic performance, the ROC curve, and the optimal threshold. The analysis therefore requires an explicit model specifying how measurement uncertainty enters the measurements, its magnitude, and its dependence across measurements. It must also distinguish distributions approximating measurements with negligible measurement uncertainty from observed distributions that already include its contribution, so that this contribution is not counted twice. [8–10].

1.3.Measurement distributions

Each population is modeled independently as normal, lognormal, or gamma using arithmetic-scale means and standard deviations. The selected family determines the distribution tails and the third and fourth central moments used in sampling covariance, affecting both point estimates and uncertainty [11–13].

1.4.Previous research

Parametric ROC analysis and analytical measurement uncertainty offer complementary perspectives on diagnostic performance [8–10, 13–16]. Threshold-specific uncertainty analysis connects these perspectives through explicitly defined primitive inputs and dependence assumptions.

Previous tools examined relationships between diagnostic accuracy and measurement uncertainty, diagnostic-measure uncertainty, and Bayesian diagnostic probabilities [17–19]. The present framework uses an implementation that explicitly accounts for covariance among the input estimates. The framework retains within-population mean–standard deviation covariance and measurement covariance between sensitivity and specificity. Its source decomposition follows the covariance form of the law of propagation of uncertainty [17–19].

2. Methods

2.1.Study inputs and diagnostic performance

We modeled test measurements separately in diseased and nondiseased groups, using a normal, lognormal, or gamma distribution for each group. The inputs were each group’s sample size, arithmetic mean, and standard deviation, together with disease prevalence, the specified diagnostic threshold, and measurement standard uncertainty evaluated at that threshold. A result at or above the threshold was classified as positive. The fitted distributions gave Se and Sp; these values and prevalence were then used to calculate the sixteen measures listed in Table 2.

2.2.Sources of uncertainty

We included sampling uncertainty in population parameters, sampling uncertainty in prevalence when applicable, and propagated measurement uncertainty [6]. Within each population, the calculations retained the covariance between the estimated mean and standard deviation. At the fixed diagnostic threshold, joint propagation also retained the covariance between Se and Sp arising from measurement uncertainty. Under the assumed measurement model, their opposite measurement sensitivity coefficients produce a negative covariance contribution, which was included when calculating uncertainty in measures that combine both rates.

In the illustrative analysis, the fitted population distributions were adjusted using the mean measurement uncertainty to approximate distributions of measurements with negligible measurement uncertainty. Measurement uncertainty was then propagated separately through the diagnostic accuracy calculations. This adjustment does not establish the measurement covariance structure assumed in propagation, which requires separate empirical justification. Uncertainty associated with distribution fitting and the adjustment was not propagated separately.

2.3. Prevalence

For the illustrative cohort calculation, prevalence was the proportion classified as diseased, with binomial sampling uncertainty. In a case-control study with deliberately fixed group sizes, that proportion need not represent clinical prevalence.

2.4. Uncertainty estimates and intervals

At each specified threshold, uncertainty was propagated using a first-order approximation, yielding source-specific contributions, the combined standard uncertainty, and an approximate pointwise expanded-uncertainty interval at a specified nominal coverage level [6, 20, 21]. Standard uncertainty is expressed on a standard-deviation scale; the interval gives a wider range around the estimate. Intervals depend on the fitted model and may be less reliable near the limits of a measure or when a ratio has a very small denominator. Restricting an interval to its admissible range does not establish its coverage.

2.5. Threshold comparison and implementation

We evaluated performance and uncertainty across a threshold range. For the optimization analysis, the software selected the threshold that maximized each chosen measure above the nondiseased mean. At that selected threshold, it used nonlinear propagation for the common measurement uncertainty. These intervals quantify uncertainty in the diagnostic accuracy measure conditional on the estimated optimal threshold; they exclude optimal threshold uncertainty arising from estimation and selection. A clinically preferred threshold also depends on the consequences of false-positive and false-negative results.

Calculations were performed in the DiagAccU, Version 2.0.1, Wolfram Language notebook. The case-study inputs and settings are reported in Tables 3 and 4. Appendix A gives the formulas, assumptions, derivatives, uncertainty calculations, and general calculation procedures. Appendix B lists notation and software information.

3. Results

3.1. Illustrative case study

The illustrative analysis uses fasting plasma glucose (FPG) data from 414 participants aged 65–67 years in NHANES 2005–2016 [27]. Reference classification by 2-h oral glucose tolerance test (OGTT) identified 52 participants with values of at least 200 mg/dL (diseased) and 362 with values below 200 mg/dL (nondiseased). The analysis is unweighted and does not provide a population estimate.

The reference-positive fraction and its binomial standard uncertainty are:

$$v = \frac{52}{414} = 0.125603865, \quad u_v = 0.016287538$$

Approximate lognormal distributions representing measurements with negligible measurement uncertainty were obtained by adjusting each maximum-likelihood fitted variance using the mean measurement uncertainty. Table 3 gives the rounded, approximate parameters used in the uncertainty calculations.

The lognormal distributions specified by the rounded arithmetic means and standard deviations are:

$$X_D \sim \text{Lognormal}(4.878, 0.264)$$

$$X_H \sim \text{Lognormal}(4.609, 0.109)$$

The measurement standard uncertainty, evaluated at the specified diagnostic threshold, is modeled as:

$$u_{\text{meas}} = \sqrt{0.812^2 + (0.0119t)^2}$$

The measurement variance is propagated jointly through both classification rates. This contribution is distinct from the measurement uncertainty addressed by the variance adjustment. The assumed measurement covariance structure requires empirical justification; total QC variation alone does not establish it.

At the clinical threshold of 126 mg/dL, the measurement standard uncertainty is $u_{\text{meas}}(126) = 1.705$ mg/dL.

Table 3. Rounded population inputs and lognormal summaries for the illustrative example. Concentrations are in mg/dL.

Analysis input or model quantity	Diseased	Nondiseased
Sample size	52	362
Analysis mean (mg/dL)	136.0	101.0
Analysis SD (mg/dL)	36.6	11.0
Model median (mg/dL)	131.327	100.406
Model skewness	0.827	0.328
Model kurtosis (not excess)	4.240	3.192

Table 4. Calculation settings for the illustrative example.

Setting	Value and scope
Population families	Lognormal–lognormal; alternative distribution families are compared in Figure 9.
Sample sizes	$n_D = 52; n_H = 362$
Arithmetic means	$m_D = 136.0; m_H = 101.0$ mg/dL
Arithmetic SDs	$s_D = 36.6; s_H = 11.0$ mg/dL
Threshold	$t = 126$ mg/dL for Figures 1, 4, 9, 10; 91–173 mg/dL for Figures 3, 5–8
Prevalence	$v = 52/414$; varied from 0.001 to 0.999 only in Figure 4
Measurement uncertainty coefficients	$b_0 = 0.812$ mg/dL; $b_1 = 0.0119$
Sampling degrees of freedom	$\nu_D = 51; \nu_H = 361; \nu_v = 413$
Measurement-source degrees of freedom	$\nu_{\text{meas}} = 1349$
Intervals	Nominal two-sided 95%; Coverage factor derived from Student's t distribution using effective degrees of freedom calculated from the contributions of the independent uncertainty sources.; admissible-range restriction

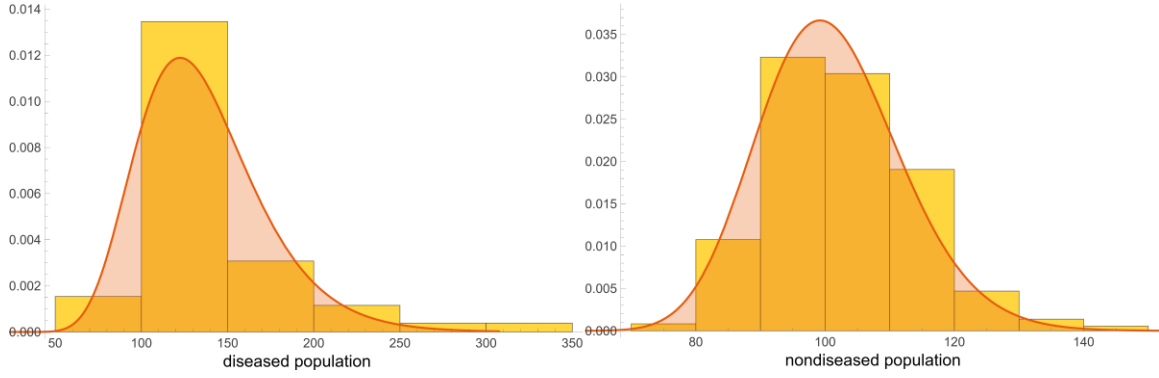


Figure 2. FPG histograms (yellow) and approximate lognormal distributions (orange) for 414 NHANES participants aged 65–67 years: 52 with 2-h OGTT values of at least 200 mg/dL (diseased) and 362 with values below 200 mg/dL (nondiseased). FPG is in mg/dL.

Figure 1 illustrates classification at 126 mg/dL, with FPG values at or above this threshold classified as positive.

3.2. Program application

The measures and their uncertainties are evaluated over 91–173 mg/dL using the settings in Table 4. Figure 4 examines prevalence effects at a fixed threshold.

3.2.1. Threshold-dependent profiles

Increasing the threshold reduces Se and increases Sp and, generally, PPV. NPV exhibits a shallow maximum before declining, whereas ACC reaches a maximum near 130 mg/dL. The measures are therefore maximized at different thresholds (Figure 3).

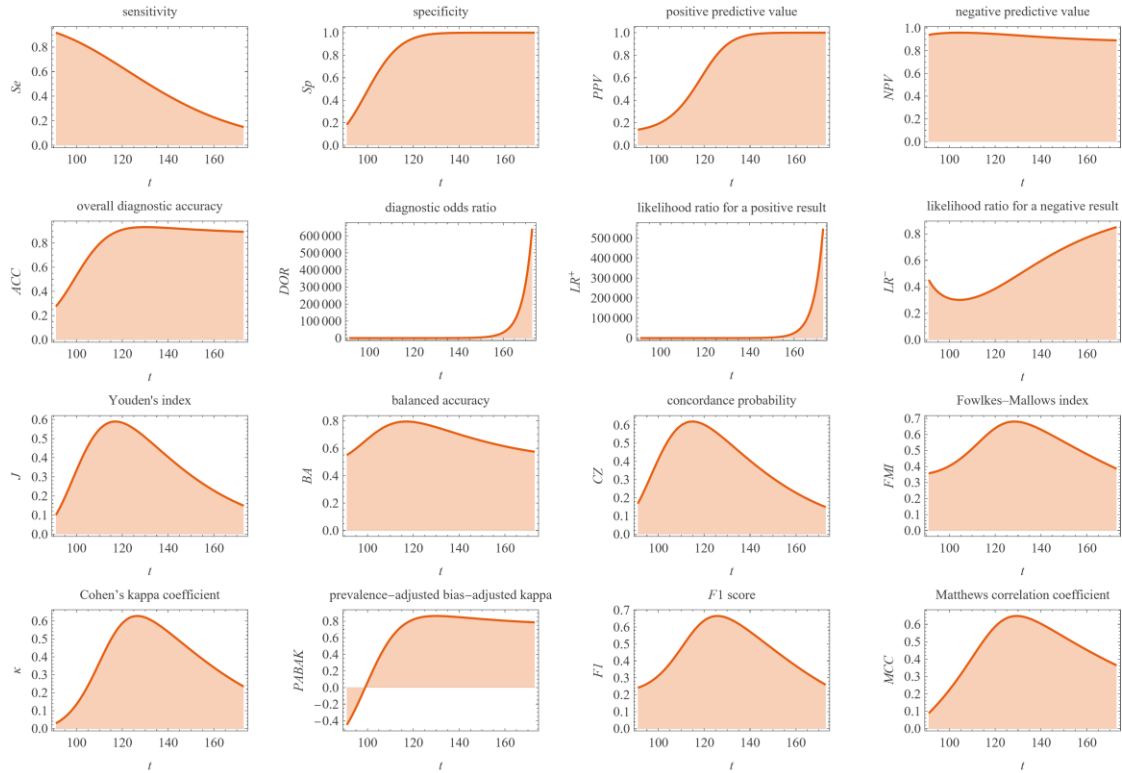


Figure 3. Diagnostic accuracy across thresholds

Profiles of the 16 diagnostic accuracy measures over thresholds of 91–173 mg/dL, using the population inputs and measurement-uncertainty settings in Tables 3 and 4.

3.2.2. Prevalence-dependent profiles

At 126 mg/dL, increasing prevalence raises PPV and lowers NPV. ACC and PABAK decrease, whereas FMI and F_1 increase. κ and MCC attain interior maxima (Figure 4).

The profiles show point estimates with fixed population distributions; prevalence-independent measures are omitted.

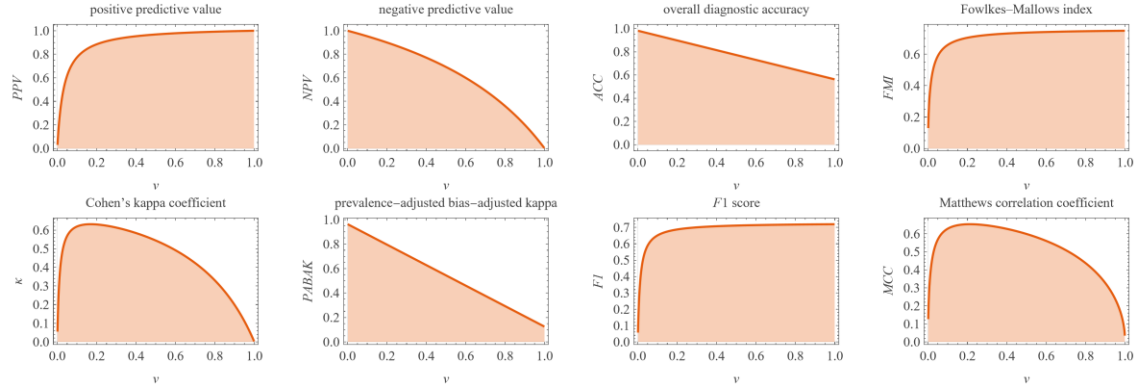


Figure 4. Diagnostic accuracy across prevalence values

Profiles of eight diagnostic accuracy measures over prevalence values of 0.001–0.999, with the FPG threshold fixed at 126 mg/dL and the population distributions held constant.

3.2.3. Relations among measures

Figure 5 relates selected pairs of measures evaluated at a common threshold. Direction reversals indicate that one value of a measure may correspond to multiple values of another.

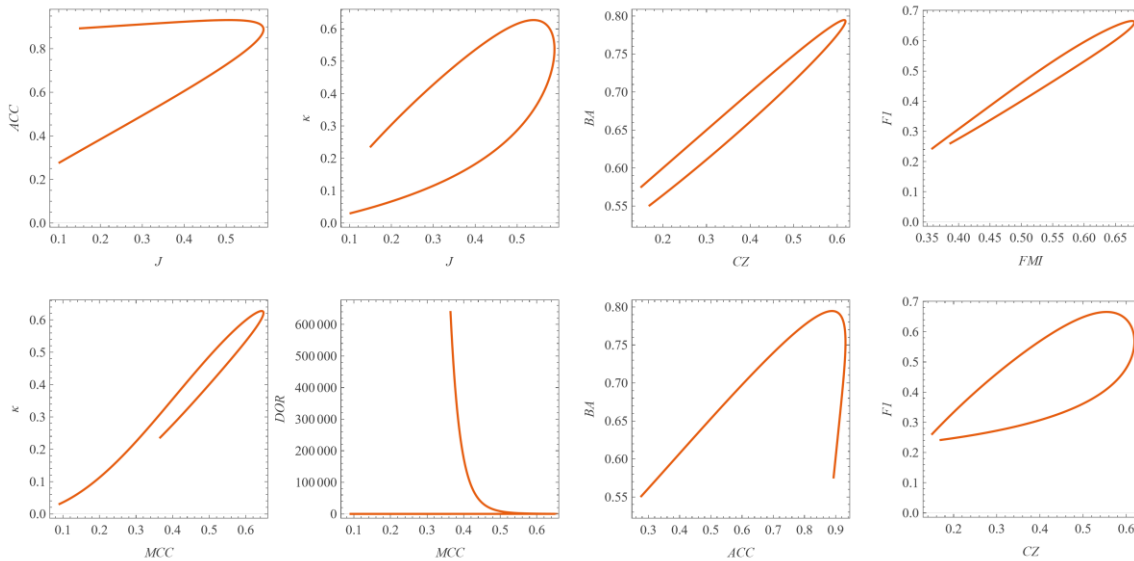


Figure 5. Relations among diagnostic accuracy measures

Relations among diagnostic accuracy measures over thresholds of 91–173 mg/dL. Each curve pairs measures evaluated at a common threshold.

3.2.4. Uncertainty

Figure 6 presents total sampling, measurement, and combined standard uncertainties, with covariance retained in the propagation.

At 126 mg/dL, PPV has total sampling and measurement standard uncertainties of 0.0441 and 0.0406, respectively, and a combined standard uncertainty of 0.0600.

Sampling uncertainty dominates many measures near 126 mg/dL, whereas measurement uncertainty predominates for Sp and LR⁺. Opposite changes in Se and Sp can reduce the first-order measurement contribution.

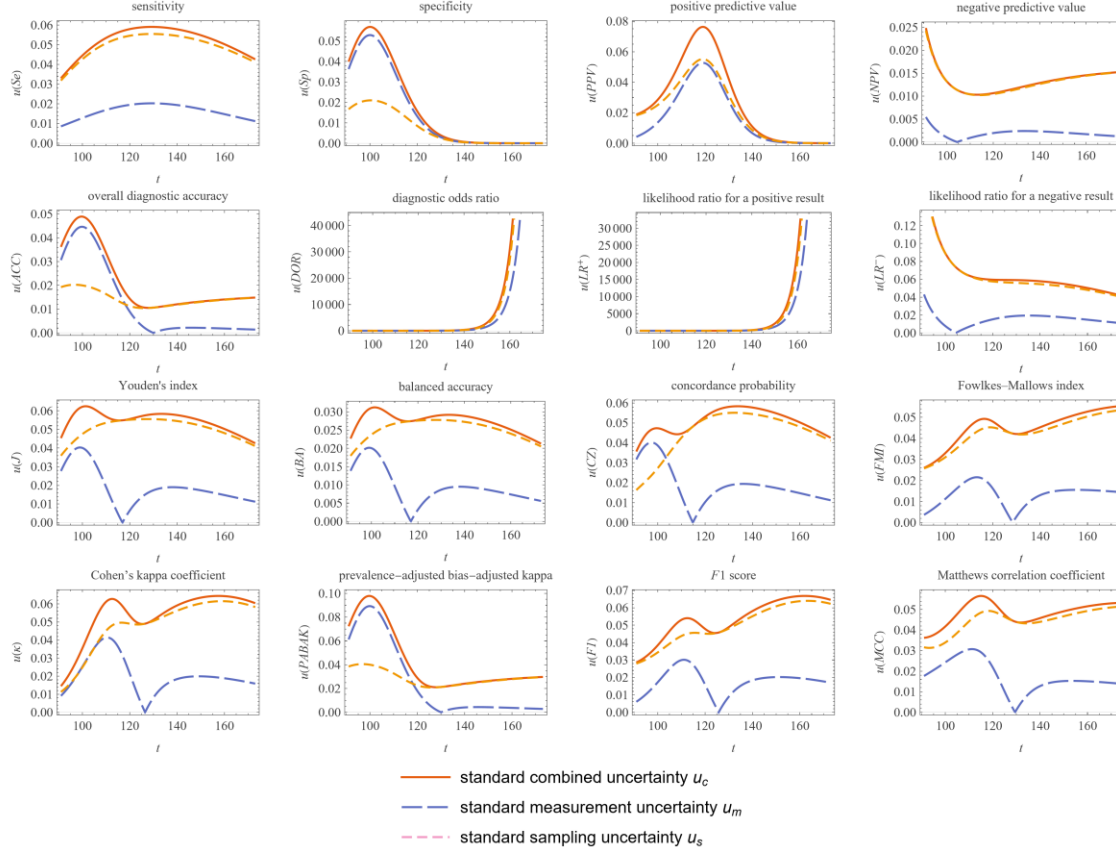


Figure 6. Sources of standard uncertainty across thresholds

Combined (solid orange), measurement (long-dashed blue), and total sampling (short-dashed) standard uncertainties across thresholds. Sampling includes contributions from both populations and from prevalence when included; covariance is retained within each source. The measurement component shown in the software output is the common measurement uncertainty contribution defined here.

3.2.5. Relative uncertainty

Relative standard uncertainty is the standard uncertainty divided by the absolute measure value. It can become very large near zero and is undefined at zero; neither feature independently indicates a change in diagnostic performance.

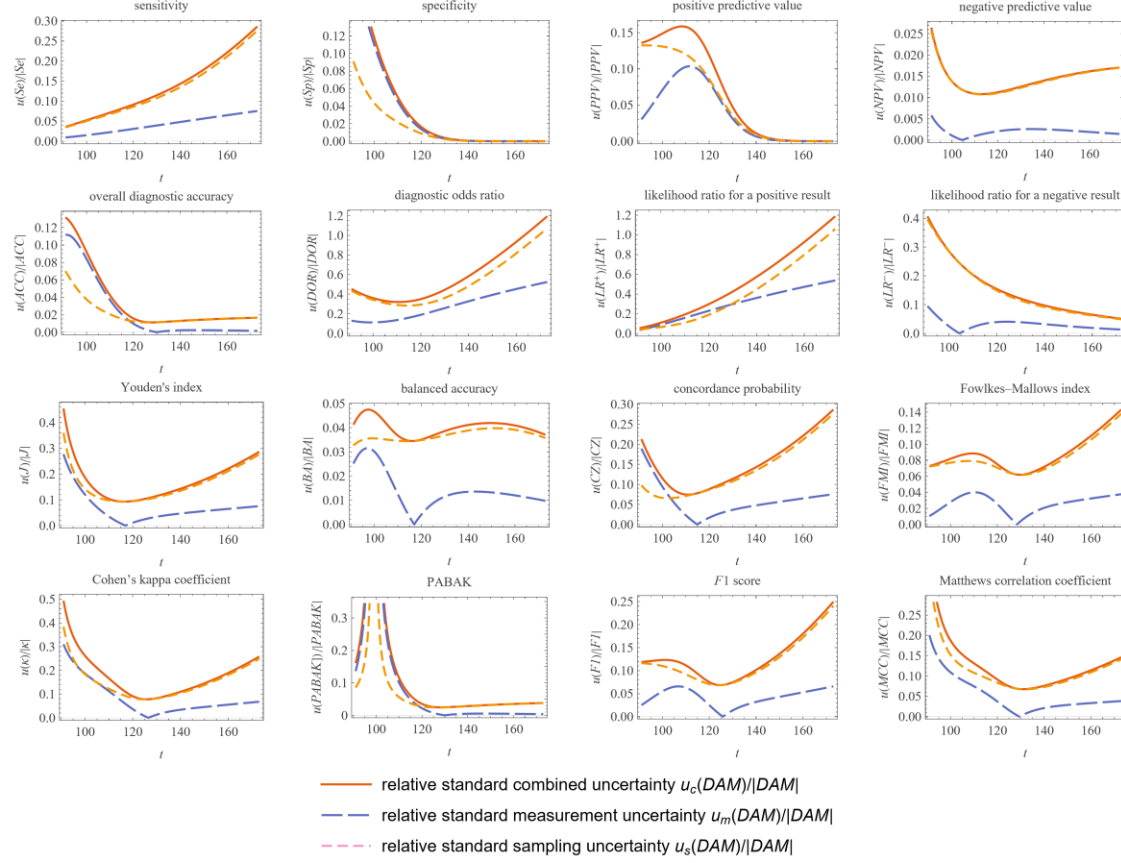


Figure 7. Relative standard uncertainty across thresholds

Relative standard uncertainties of the diagnostic accuracy measures, using the component styles in Figure 6. Relative uncertainty is undefined at zero measure values.

3.2.6. Pointwise uncertainty intervals

Figure 8 shows approximate pointwise 95% expanded-uncertainty intervals calculated using effective degrees of freedom derived from the source-specific variance contributions. Interval limits are restricted to each measure's admissible range. The nominal 95% level refers to each threshold individually; these intervals do not constitute simultaneous confidence bands across thresholds.

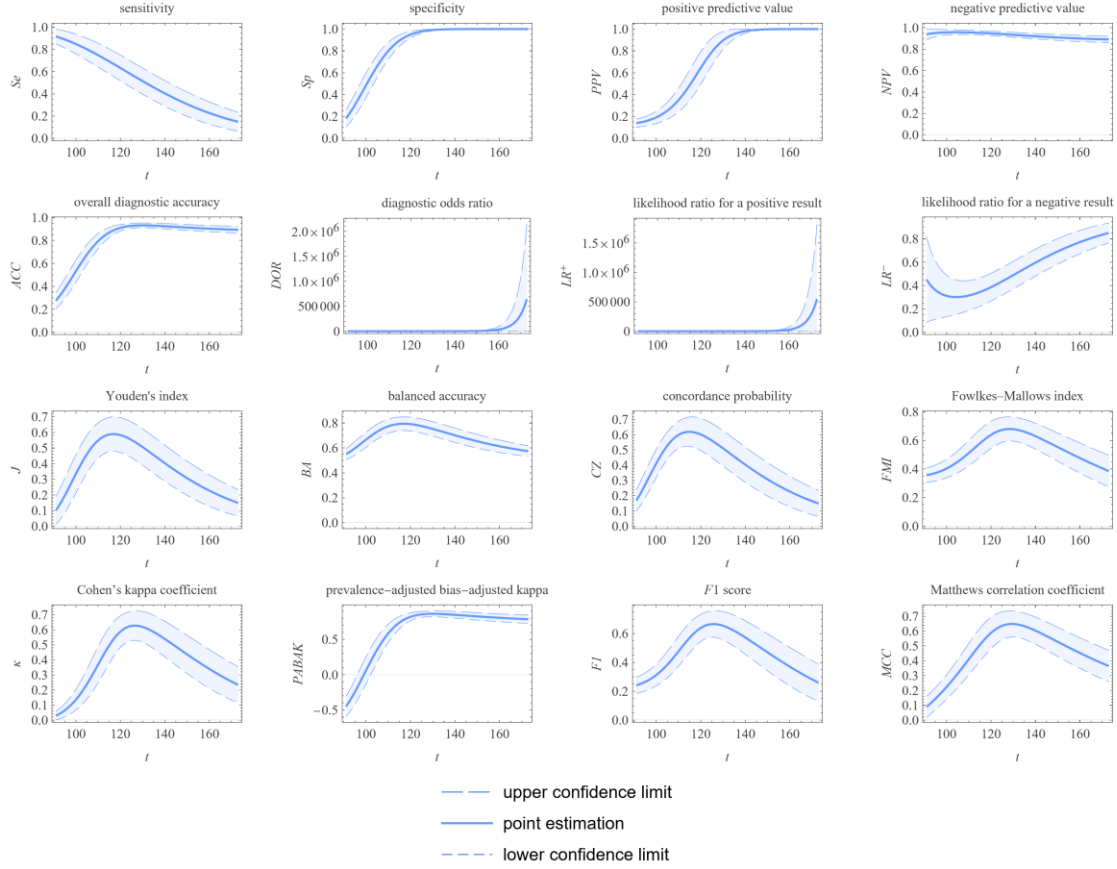


Figure 8. Pointwise expanded-uncertainty intervals across thresholds

Point estimates (curves) and approximate pointwise 95% expanded-uncertainty intervals (shading and dashed interval limits) across thresholds.

3.2.7. Estimates and intervals at the clinical threshold

At 126 mg/dL, PPV is 0.8155, with standard uncertainty 0.0600 and an approximate 95% expanded-uncertainty interval of 0.6980–0.9331. NPV is 0.9398, with standard uncertainty 0.0113 and an approximate 95% expanded-uncertainty interval of 0.9176–0.9620. Figure 9 compares distribution families; Figure 10 presents estimates and intervals for all 16 measures under the lognormal model.

point estimates of diagnostic accuracy measures																	
prevalence of disease $\nu = 0.126$																	
measurement distribution		Se	Sp	ACC	PPV	NPV	DOR	LR ⁺	LR ⁻	J	BA	CZ	FMI	κ	PABAK	F ₁	MCC
diseased	nondiseased																
normal	normal	0.608	0.988	0.941	0.883	0.946	132.880	52.742	0.397	0.596	0.798	0.601	0.733	0.688	0.881	0.720	0.703
	lognormal	0.608	0.982	0.935	0.827	0.946	83.244	33.268	0.400	0.589	0.795	0.597	0.709	0.665	0.869	0.701	0.675
	gamma	0.608	0.984	0.937	0.845	0.946	95.146	37.937	0.399	0.592	0.796	0.598	0.717	0.673	0.873	0.707	0.684
lognormal	normal	0.562	0.988	0.935	0.875	0.940	110.190	48.799	0.443	0.551	0.775	0.556	0.701	0.650	0.870	0.685	0.670
	lognormal	0.562	0.982	0.929	0.816	0.940	69.026	30.780	0.446	0.544	0.772	0.552	0.677	0.627	0.858	0.666	0.641
	gamma	0.562	0.984	0.931	0.834	0.940	78.895	35.101	0.445	0.546	0.773	0.553	0.685	0.635	0.862	0.672	0.650
gamma	normal	0.575	0.988	0.937	0.878	0.942	116.210	49.931	0.430	0.564	0.782	0.569	0.711	0.661	0.873	0.695	0.680
	lognormal	0.575	0.982	0.931	0.819	0.941	72.797	31.494	0.433	0.557	0.779	0.565	0.686	0.638	0.861	0.676	0.651
	gamma	0.575	0.984	0.933	0.838	0.942	83.206	35.915	0.432	0.559	0.780	0.566	0.694	0.646	0.865	0.682	0.660

Figure 9. Diagnostic accuracy under alternative distribution families

Diagnostic accuracy estimates at 126 mg/dL for nine combinations of normal, lognormal, and gamma population distributions, with arithmetic means, standard deviations, sample sizes, and prevalence held fixed.

95.00% confidence intervals			
prevalence of disease $\nu = 0.126$			
measure	point estimate	lower confidence limit	upper confidence limit
Se	0.562	0.445	0.680
Sp	0.982	0.968	0.996
ACC	0.929	0.908	0.950
PPV	0.816	0.698	0.933
NPV	0.940	0.918	0.962
DOR	69.026	14.805	123.250
LR ⁺	30.780	8.439	53.121
LR ⁻	0.446	0.327	0.564
<i>J</i>	0.544	0.429	0.659
BA	0.772	0.715	0.829
CZ	0.552	0.438	0.666
FMI	0.677	0.591	0.764
κ	0.627	0.530	0.725
PABAK	0.858	0.816	0.900
F_1	0.666	0.574	0.757
MCC	0.641	0.548	0.734

Figure 10. Diagnostic accuracy and uncertainty at the clinical threshold

Point estimates and approximate pointwise 95% expanded-uncertainty intervals for the 16 diagnostic accuracy measures at 126 mg/dL under the lognormal model.

3.2.8. Optimal diagnostic thresholds

CZ is maximized at 114.859 mg/dL, *J* and BA at 116.948 mg/dL, and ACC and PABAK at 130.011 mg/dL. Although the first-order threshold derivative vanishes at a smooth optimum, nonlinear propagation retains a measurement contribution. For ACC, the measurement and combined standard uncertainties are 0.000568 and 0.010659, respectively. The intervals are conditional on the estimated optimal threshold and exclude optimal threshold uncertainty arising from repeated estimation and selection. Figure 11 reports the current calculations for the nine measures with the reported optima.

95.00% confidence intervals				
prevalence of disease $v = 0.126$				
measure	optimal threshold t	point estimate	lower confidence limit	upper confidence limit
ACC	130.011	0.932	0.911	0.953
J	116.948	0.589	0.479	0.700
BA	116.948	0.795	0.740	0.850
CZ	114.859	0.619	0.522	0.716
FMI	128.416	0.680	0.596	0.765
κ	126.529	0.628	0.529	0.726
PABAK	130.011	0.863	0.821	0.905
F_1	125.753	0.666	0.574	0.757
MCC	129.421	0.648	0.560	0.737

Figure 11. Diagnostic accuracy measures at estimated optimal thresholds, with conditional approximate 95% expanded-uncertainty intervals.

The displayed confidence limits are approximate 95% expanded-uncertainty limits. Common measurement uncertainty was propagated by nine-point Gauss–Hermite quadrature under the model. The intervals are conditional on the estimated optimal threshold and exclude optimal threshold uncertainty.

3.2.9. Effect of including covariance terms

At 126 mg/dL, omitting within-population mean–standard deviation covariance decreased the combined standard uncertainty of PPV but increased those of NPV and ACC (Table 5). Omitting only the measurement-induced covariance increased the measurement contribution for each of these measures. For ACC, it increased from 0.002351 to 0.005502. Dependence can therefore change both the magnitude of uncertainty and the interpretation of its sources; omitting covariance does not provide a uniformly conservative calculation.

Table 5. Effects of covariance omission at the illustrative threshold.

Measure	Combined uncertainty	Combined uncertainty without mean–SD covariance	Measurement uncertainty	Measurement uncertainty without induced covariance
PPV	0.059962	0.059031	0.040610	0.046321
NPV	0.011265	0.012075	0.002284	0.002626
ACC	0.010699	0.011409	0.002351	0.005502

All entries are dimensionless standard uncertainties at the inputs in Tables 3 and 4. The third column removes only the two within-population mean–SD covariance terms. The fifth column sets the measurement-induced sensitivity–specificity covariance to zero while retaining both marginal measurement variance contributions. These are controlled changes to the propagation model, not alternative empirically established measurement mechanisms. Point estimates are unchanged. *Supplemental file 1* describes the comparisons.

4. Discussion

4.1. Relationship to Bayesian Diagnostic Insights

The previously published software *Bayesian Diagnostic Insights* compares predictive values with measurement-specific probabilities of disease and nondisease [19]. It supports all nine ordered combinations of normal, lognormal, and gamma populations and calculates measurement, sampling, and total standard uncertainties. These capabilities are shared with *DiagAccU*.

DiagAccU uses distribution-specific third and fourth central moments to calculate the covariance between the estimated mean and standard deviation within each population. The normal–lognormal positive predictive value expression examined in *Bayesian Diagnostic Insights* includes separate variance contributions from the estimated means and standard deviations but omits their within-population covariance terms. *DiagAccU* calculates sixteen diagnostic accuracy measures and their joint covariance matrices, with separate contributions from each uncertainty source.

Both implementations evaluate the population distribution tails at a common diagnostic threshold. *DiagAccU* propagates the measurement variance through the complete measure expressions, retaining the specified covariances while keeping the diagnostic threshold fixed. The controlled comparisons in Table 5 quantify the effects of selected covariance terms within *DiagAccU*; they do not constitute a numerical comparison of the two applications.

4.2. Principal findings

The framework propagates uncertainty for sixteen measures across nine population-family pairs. The illustrative PPV and NPV estimates show how sampling uncertainty and common measurement uncertainty contribute to the resulting intervals.

Maximizing a point estimate need not minimize combined or relative uncertainty. At a smooth optimum, the first-order threshold derivative may vanish while nonlinear propagation of measurement uncertainty still contributes to uncertainty. Neither a zero first derivative nor a high estimate establishes clinical preference.

4.3. Relations between measures

The sixteen outputs depend on Se, Sp, and prevalence, so their first-order covariance has rank at most three. They provide complementary interpretations rather than sixteen independent assessments. The identities relating BA to J and PABAK to ACC explain their common maximizing thresholds and proportional uncertainties.

4.4. Uncertainty decomposition

Including covariance terms for each uncertainty source affects the resulting variance. For PPV, accounting for covariance between the estimated mean and standard deviation decreases the variance contribution from the diseased population and increases the corresponding contribution from the nondiseased population. The measurement contribution combines both tail derivatives before squaring, accounting for the single measurement-uncertainty input.

Excluding prevalence uncertainty changes the combined standard uncertainty of PPV from 0.059962 to 0.055657 and that of NPV from 0.011265 to 0.007517, with unchanged point estimates. The sampling interpretation determines whether this source is included; the smaller uncertainty is not a selection criterion.

4.5. Additional sources of uncertainty and bias

Reference classification: Group labels are treated as fixed. OGTT misclassification, reference-standard uncertainty, verification bias, and incomplete disease ascertainment are excluded [23].

Distributional specification: Normal, lognormal, and gamma models may inadequately represent mixtures, truncation, censoring, or heavy tails. Checks across the nine family pairs verify executable support, not fit to participant data [11–13, 24].

Prevalence and transportability: Modeled-cohort uncertainty assumes a binomial count ratio independent of the fitted summaries. Fixed strata, survey sampling, and deployment populations may require different prevalence inputs and additional covariance terms.

Behavior near parameter limits: First-order intervals may perform poorly near zero denominators or parameter limits. Transformed, likelihood-based, joint Monte Carlo, and bootstrap alternatives require separate assumptions and validation; admissible-range restriction alone is insufficient [25–28].

Sampling approximations: The sample-mean variance is exact under the stated independent-sample model with finite variance; covariance terms involving the sample standard deviation are approximated to first order. Normality does not make the propagated diagnostic accuracy measure uncertainty or interval exact.

Optimal threshold uncertainty: Estimating and selecting an optimal threshold from fitted inputs introduces uncertainty and dependence omitted by the conditional intervals reported here. Resampling should repeat model estimation and threshold selection in each replicate [29–31].

Transfer of the measurement-uncertainty model: Coefficients b_0 and b_1 are treated as known and shared by both populations. Propagating their estimation covariance, matrix or instrument changes, or correlations with fitted summaries requires an expanded input model.

Dependence: The model includes within-population mean–standard deviation covariance but excludes cross-population covariance, clustering, repeated measurements, and dependence between prevalence and fitted summaries. The common measurement uncertainty input is assumed to be independent of fitted summaries.

Decision rule and intervals: Reversing the positive-test orientation requires redefining both tails. Pointwise intervals across measures and thresholds are not simultaneous intervals; admissible-range restriction does not establish nominal coverage.

4.6. Implementation strengths

Common parameterization: Arithmetic means and standard deviations retain the same interpretation across all three population families.

Uncertainty components: The uncertainty of each measure includes contributions from both populations, prevalence when applicable, and the common measurement uncertainty input, except where contributions are zero by definition or explicitly excluded.

Consistency of individual and joint variance calculations: For first-order propagation at a specified threshold, the joint covariance diagonal reproduces scalar combined variances to numerical precision, and the source covariance matrices sum to the total. The conditional nonlinear calculation at optimized thresholds is described separately in *Supplemental file 1, Section A.10*.

Reproducibility: The source notebook implements the calculations; the companion notebook and *Supplemental file 1* document their mathematical structure. Tables 3 and 4 define the numerical example. The case-study input derivation and the sampling properties of the fitted and adjusted estimators are separate from reproduction of the conditional calculations.

4.7. Contribution

DiagAccU calculates uncertainty for sixteen threshold-dependent measures, including covariance terms. Its joint covariance matrices and decomposition of uncertainty by source allow researchers to compare diagnostic accuracy measures while accounting for their dependence and identify the sources that limit precision. The controlled comparisons demonstrate that dependence assumptions can change reported precision and the contributions of individual uncertainty sources. Clinically, these analyses can inform appraisal of the precision of evidence supporting a test's proposed role in ruling disease in or out, and help prioritize further data collection or improvements in measurement. The software therefore supports more transparent evaluation of diagnostic performance and its uncertainty. Selecting a preferred clinical threshold additionally requires a defined target population, its disease prevalence, and an assessment of the consequences of false-positive and false-negative classifications within the intended care pathway.

DiagAccU calculates uncertainty for sixteen threshold-dependent measures, including covariance terms. Its joint covariance matrices and decomposition of uncertainty by source allow researchers to compare diagnostic accuracy measures while accounting for their dependence and identify the sources that limit precision. The controlled comparisons demonstrate that dependence assumptions can change reported precision and the contributions of individual uncertainty sources. Clinically, these analyses can inform appraisal of the precision of evidence supporting a test's proposed role in ruling disease in or out, and help prioritize further data collection or improvements in measurement. By quantifying the contribution of measurement uncertainty to uncertainty in diagnostic performance at specified thresholds, *DiagAccU* can also inform the definition of analytical performance specifications for clinical laboratories, thereby supporting quality planning and risk management in relation to the test's intended clinical use [32, 33]. Selecting a preferred clinical threshold additionally requires a defined target population, its disease prevalence, and an

assessment of the consequences of false-positive and false-negative classifications within the intended care pathway.

4.8. Reporting recommendations

- a) Specify the positive-test orientation, population families, arithmetic summaries, sample sizes, and prevalence interpretation.
- b) Report point estimates, source-specific and combined standard uncertainties, effective degrees of freedom, coverage factors, interval levels, and any endpoint restriction.
- c) Distinguish intervals at a specified diagnostic threshold from intervals conditional on an estimated optimum. Assess optimal threshold uncertainty through a specified joint simulation or resampling analysis that repeats estimation and selection of the optimum.
- d) Specify the function for measurement standard uncertainty, the coefficient values and their units, how uncertainty in the coefficients is treated, and the degrees of freedom assigned to measurement uncertainty.

Limitations and future work

The illustrative analysis uses rounded lognormal summaries adjusted for analytical variation and approximate covariance formulas, without separately propagating uncertainty from fitting or variance adjustment; measurement-uncertainty coefficients are treated as fixed. Results depend on distributional assumptions, first-order propagation, and approximate effective degrees of freedom. Intervals at estimated optimal thresholds exclude threshold-selection uncertainty. Controlled comparisons demonstrate covariance effects but do not establish interval coverage, which requires simulation and resampling that repeat fitting and threshold selection. Further extensions should incorporate uncertainty and covariance in measurement-uncertainty coefficients, survey design, and reference-classification uncertainty [31, 34–39]. Prevalence cannot be specified independently of population counts, limiting application to prevalence-dependent measures in case-control studies and populations with a different prevalence.

5. Conclusions

DiagAccU calculates uncertainty for sixteen threshold-dependent diagnostic accuracy measures under independently specified normal, lognormal, or gamma population models. It incorporates population-summary and measurement-induced covariance, reports joint output covariance, and supports conditional nonlinear propagation at estimated optimal thresholds.

The illustrative calculations show that covariance assumptions affect both the magnitude of uncertainty and its attribution to individual sources. Joint covariance reporting makes dependence among diagnostic accuracy measures explicit, supporting comparisons that account for their shared sources of uncertainty. Numerical checks and controlled comparisons support the implemented calculations within the specified models.

By quantifying the contribution of measurement uncertainty to uncertainty in diagnostic performance, *DiagAccU* can inform the definition of analytical performance specifications for clinical laboratories and support quality planning and risk management in relation to intended clinical use. Clinical interpretation additionally requires a defined target population and consideration of the consequences of false-positive and false-negative classifications.

The software supports research comparisons, model appraisal, and uncertainty reporting. Its intervals remain approximate, and results at estimated optimal thresholds exclude threshold-selection uncertainty. Further evaluation should assess interval coverage through simulation and resampling that repeat distribution fitting and threshold selection. Improved clinical decisions or patient outcomes have not been established.

6. Supplemental material

The following supplemental files are available for download as a ZIP archive at:

<https://www.hcsl.com/Supplements/SDAMUR.zip> (accessed on September 21, 2026):

- a) Supplemental File I:
DiagAccU.nb: The program as a Wolfram Mathematica Notebook.
- b) Supplemental File II:
DiagAccUCalculations.nb: The calculations for the estimation of the DAMs and their combined standard uncertainty in a Wolfram Mathematica Notebook.
- c) Supplemental File III:
DiagAccUInterface.pdf: A brief documentation of the interface of the program.

7. Declarations

Author Contributions: Conceptualization: RCA and TC; methodology: RCA and ATH; software: RCA, TC and ATH; validation: TC; formal analysis: RCA and ATH; investigation: TC; resources: ATH; data curation: TC; writing—original draft preparation: RCA; writing—review and editing: ATH; visualization: RCA and TC; supervision: ATH; project administration: RCA. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Data collection followed the Declaration of Helsinki. The National Center for Health Statistics Ethics Review Board approved data collection and online release for public use (Protocols #2005-06 and #2011-17). See <https://www.cdc.gov/nchs/nhanes/about/erb.html> (accessed on September 17, 2026).

Informed Consent Statement: Written consent was obtained from each survey participant.

Consent for publication: Not applicable.

Data Availability Statement: NHANES 2005–2016 participant data are publicly available from the National Center for Health Statistics [27]. Tables 3 and 4 provide the approximate population inputs and settings used in the illustrative calculations.

Conflicts of Interest: The authors declare no conflicts of interest.

8. Acknowledgements

OpenAI Codex assisted with manuscript restructuring, language editing, and preparation of the supplementary documents.

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10. Citation

Chatzimichail RA, Chatzimichail T, Hatjimihail AT. Uncertainty estimation of diagnostic accuracy measures under parametric distributions. Technical Report XXIX. Hellenic Complex Systems Laboratory; 2025. DOI: <https://doi.org/10.5281/zenodo.22963267>

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Appendix A. Mathematical and statistical calculations

This appendix provides the mathematical and statistical specification from input definition to expanded-uncertainty intervals. Section A.6 gives the complete derivative table, A.8 gives the joint covariance formulation, and A.11 defines the implementation checks and controlled covariance comparisons. It documents *DiagAccU.nb*; *Bayesian Diagnostic Insights* is the separate software underlying the earlier publication [19].

A.1. Variance, covariance, and first-order uncertainty propagation

Variance is the square of the corresponding standard uncertainty. Covariance describes how two uncertain quantities vary together. A partial derivative describes the local rate of change of an output with respect to one input, with all other inputs held fixed. A gradient collects the partial derivatives for one output, and a Jacobian collects them for several outputs. First-order uncertainty propagation is based on a local linear approximation.

A.2. Input specification and ordering

The population labels D and H denote diseased and nondiseased populations, respectively. Prevalence is denoted v , and the intermediate arguments a and b denote sensitivity (Se) and specificity (Sp). A generic diagnostic accuracy measure is denoted F . The standard uncertainties u_c , u_s , and u_m , evaluated for a specified measure, denote combined, total sampling, and measurement uncertainty, respectively. The measurement component is obtained by joint propagation of measurement uncertainty through both classification rates. The population-parameter and prevalence sampling components are denoted u_{par} and u_{prev} . Superscript T denotes transpose. The measure is a function of the inputs, $F = g(\mathbf{x})$.

Scalar variables are italic; measure abbreviations, named operators, and descriptive subscripts are upright. Vectors and matrices are bold. The input Jacobian is denoted $\mathbf{M}$, and the four source-specific variance contributions are denoted V_D , V_H , V_v , and V_{meas} . These correspond to the Jacobian $\mathbf{J}$ and the variance contributions V_D , V_H , V_v , and V_{meas} , respectively, in Appendix B. The symbol J denotes Youden's index throughout this appendix.

The diagnostic threshold t is a specified decision value and is fixed at each evaluation. For numerical propagation, τ denotes the common argument of the two cumulative distribution functions, with nominal value $\tau = t$. The input covariance matrix assigns the measurement variance to this computational coordinate; no uncertainty is assigned to the diagnostic threshold. The input vector comprises the four fitted arithmetic summaries, prevalence, and the evaluation coordinate. The intermediate vector comprises the three arguments of each diagnostic accuracy measure.

$$\mathbf{x} = \begin{pmatrix} m_D \\ s_D \\ m_H \\ s_H \\ v \\ \tau \end{pmatrix} \in \mathbb{R}^6, \quad (A.1)$$

$$\mathbf{z} = \begin{pmatrix} a \\ b \\ v \end{pmatrix} = \begin{pmatrix} \text{Se} \\ \text{Sp} \\ v \end{pmatrix} \in \mathbb{R}^3.$$

The input and intermediate vectors have six and three components, respectively. All vectors are column vectors. The aliases $a = \text{Se}$ and $b = \text{Sp}$ are used in the derivative formulas; v denotes prevalence throughout.

$$\mathbf{m} = \begin{pmatrix} \text{Se} \\ \text{Sp} \\ \text{ACC} \\ \text{PPV} \\ \text{NPV} \\ \text{DOR} \\ \text{LR}^+ \\ \text{LR}^- \\ J \\ \text{BA} \\ \text{CZ} \\ \text{FMI} \\ \kappa \\ \text{PABAK} \\ F_1 \\ \text{MCC} \end{pmatrix} \in \mathbb{R}^{16}. \quad (\text{A.2})$$

Equation (A.2) specifies the output order. Row i of each output Jacobian corresponds to measure i , and element (i, j) of the output covariance matrix is the covariance between measures i and j .

Observations are assumed to be independent within each population, and samples from different populations are assumed to be mutually independent. The supplied standard deviations are assumed to be the square roots of Bessel-corrected sample variances. The prevalence and measurement uncertainty sources are assumed independent of each other and of the population-parameter sampling sources. These assumptions determine which entries of the input covariance matrix are zero. For alternative study designs, the covariance matrix must represent the applicable dependence structure. Application of these formulas to fitted or analytically adjusted summaries constitutes a model-based approximation. Uncertainty arising from fitting or variance adjustment is accounted for only if explicitly incorporated into an expanded input covariance model.

A.3. Distribution fitting and calculation of the two classification rates

For population $P \in \{D, H\}$, the normal, lognormal, and gamma families are parameterized by the arithmetic mean μ_P and standard deviation σ_P . Lognormal and gamma models require positive means. Tail probabilities at thresholds outside a distribution's support follow its support conventions. The gamma parameters ξ and θ denote shape and scale, respectively.

$$\begin{aligned} X_N &\sim \mathcal{N}(\mu, \sigma^2). \\ \sigma_L^2 &= \log\left(1 + \left(\frac{\sigma}{\mu}\right)^2\right), \\ \mu_L &= \log(\mu) - \frac{\sigma_L^2}{2}. \\ X_L &\sim \text{Lognormal}(\mu_L, \sigma_L). \\ \xi &= \left(\frac{\mu}{\sigma}\right)^2, \quad \theta = \frac{\sigma^2}{\mu}, \\ X_G &\sim \text{Gamma}(\xi, \theta). \end{aligned}$$

The selected distribution family is parameterized using the supplied arithmetic mean and standard deviation. For the lognormal family, the transformed parameters describe the logarithm of the measurement. For the gamma family, the shape and scale are obtained from the two arithmetic summaries. Agreement with the arithmetic summaries does not establish adequate fit in the distribution tails.

For a positive-above-threshold test, Se is the probability in the diseased upper tail and Sp is the probability in the nondiseased lower tail. A cumulative distribution function gives the probability below a threshold. Thus Se is one minus the diseased cumulative probability, whereas Sp is the nondiseased cumulative probability.

The class-conditional cumulative distribution functions are denoted F_D and F_H . The probabilities are evaluated at the diagnostic threshold. Under the positive-above-threshold rule, Se and Sp are defined as follows:

$$\text{Se}(t) = \Pr(X_D \geq t).$$

$$\text{Sp}(t) = \Pr(X_H < t).$$

$$\text{Se}(t) = 1 - F_D(t), \quad \text{Sp}(t) = F_H(t).$$

For continuous distributions, including all three families used here, the probability of observing the threshold exactly is zero. The distinction between a strict and a non-strict inequality therefore does not alter the model probabilities.

A.4. Calculation of diagnostic accuracy measures

Tables A1 and A2 define the remaining measures. The symbols v , q , p_o , and p_e denote prevalence, predicted-positive probability, observed agreement, and chance agreement, respectively:

$$q = v \text{Se} + (1 - v) (1 - \text{Sp}).$$

$$p_o = v \text{Se} + (1 - v) \text{Sp}.$$

$$p_e = q v + (1 - q) (1 - v).$$

TABLE A1. DEFINITIONS OF Se, Sp, PPV, NPV, ACC, LR⁺, LR⁻, AND BALANCE MEASURES.

Measure	Count definition	Definition using Se, Sp, v
Se	$\frac{\text{TP}}{\text{TP} + \text{FN}}$	Se
Sp	$\frac{\text{TN}}{\text{TN} + \text{FP}}$	Sp
PPV	$\frac{\text{TP}}{\text{TP} + \text{FP}}$	$\frac{v\text{Se}}{q}$
NPV	$\frac{\text{TN}}{\text{TN} + \text{FN}}$	$\frac{(1 - v)\text{Sp}}{1 - q}$
ACC	$\frac{\text{TP} + \text{TN}}{N}$	$v\text{Se} + (1 - v)\text{Sp}$
DOR	$\frac{\text{TPTN}}{\text{FPFN}}$	$\frac{\text{Se Sp}}{(1 - \text{Se})(1 - \text{Sp})}$
LR⁺	$\frac{\frac{\text{TP}}{R}}{\frac{\text{FP}}{N - R}}$	$\frac{\text{Se}}{1 - \text{Sp}}$
LR⁻	$\frac{\frac{\text{FN}}{R}}{\frac{\text{TN}}{N - R}}$	$\frac{1 - \text{Se}}{\text{Sp}}$
J	$\frac{\text{TP}}{R} + \frac{\text{TN}}{N - R} - 1$	$\text{Se} + \text{Sp} - 1$
BA	$\frac{\frac{\text{TP}}{R} + \frac{\text{TN}}{N - R}}{2}$	$\frac{\text{Se} + \text{Sp}}{2}$
CZ	$\frac{\text{TPTN}}{R(N - R)}$	Se Sp

TABLE A2. DEFINITIONS OF ASSOCIATION AND AGREEMENT MEASURES.

Measure	Count definition	Definition using Se, Sp, v
FMI	$\frac{TP}{\sqrt{RN_+}}$	$\sqrt{Se PPV}$
κ	$\frac{N(TP + TN) - RN_+ - (N - R)(N - N_+)}{N^2 - RN_+ - (N - R)(N - N_+)}$	$\frac{p_o - p_e}{1 - p_e}$
PABAK	$2 \frac{TP + TN}{N} - 1$	$2p_o - 1$
F_1	$\frac{2TP}{2TP + FP + FN}$	$\frac{2vSe}{v + q}$
MCC	$\frac{TPTN - FPFN}{\sqrt{RN_+(N - R)(N - N_+)}}$	$\frac{\sqrt{v(1 - v)}(Se + Sp - 1)}{\sqrt{q(1 - q)}}$

In Tables A1 and A2, $N = TP + FN + FP + TN$, $R = TP + FN$, and $N_+ = TP + FP$ denote total, disease-positive marginal, and test-positive marginal counts, respectively. Dividing counts by N gives the equivalent probability definitions. Ratios and their derivatives require nonzero denominators.

A6. Classification of measures

TABLE A3. OVERLAPPING CLASSIFICATION OF THE SIXTEEN MEASURES.

Category	Measures
Agreement	κ , PABAK
Association and F-type indices	FMI, F_1 , MCC
Balance	J , BA, CZ
Bounded	Se, Sp, ACC, PPV, NPV, J , BA, CZ, FMI, κ , PABAK, F_1 , MCC
Ratio-type	DOR, LR^+ , LR^-
Prevalence invariant	Se, Sp, DOR, LR^+ , LR^- , J , BA, CZ
Prevalence dependent	ACC, PPV, NPV, FMI, κ , PABAK, F_1 , MCC

The classifications overlap: agreement measures and F-type indices may depend on prevalence. Boundedness implies neither low uncertainty nor robustness to class imbalance.

The predicted-positive probability equals the sum of the probabilities of true-positive and false-positive outcomes in the target population. PPV is the ratio of the true-positive probability to this sum. Similarly, NPV is the ratio of the true-negative probability to the sum of the true-negative and false-negative probabilities. Agreement and association measures are also calculated from these outcome probabilities.

A.5. Construction of the input covariance matrix

The input covariance matrix includes the variances of the estimated means and standard deviations for both populations, together with the prevalence variance and the measurement variance. The sixth coordinate carries the measurement variance for numerical propagation at the diagnostic threshold. Within each population, the covariance between the estimated mean and standard deviation is retained. All remaining covariances are zero under the stated independence assumptions [1–3].

A.5.1. Sampling uncertainty within each population

The centered sampling deviations are defined as $\delta_{\bar{X}} = \bar{X} - E[\bar{X}]$ and $\delta_S = S - E[S]$, where expectations are taken over repeated samples from the same population.

$$\begin{aligned}\Sigma_{\bar{X},S} &= E \left(\begin{pmatrix} \delta_{\bar{X}} \\ \delta_S \end{pmatrix} \begin{pmatrix} \delta_{\bar{X}} & \delta_S \end{pmatrix} \right), \\ \Sigma_{\bar{X},S} &= E \left(\begin{pmatrix} \delta_{\bar{X}}^2 & \delta_{\bar{X}} \delta_S \\ \delta_S \delta_{\bar{X}} & \delta_S^2 \end{pmatrix} \right), \\ \Sigma_{\bar{X},S} &\approx \begin{pmatrix} A & B \\ B & C \end{pmatrix}.\end{aligned}\tag{A.3}$$

The covariance matrix is the expectation of the outer product of the centered estimator vector with its transpose. Expectations are evaluated for each matrix element. The off-diagonal elements are equal by symmetry of covariance.

$$\begin{aligned}\Sigma_{\bar{X},S^2} &= \begin{pmatrix} \frac{\sigma^2}{n} & \frac{\mu_3}{n} \\ \frac{\mu_3}{n} & \frac{\mu_4 - \frac{n-3}{n-1}\sigma^4}{n} \end{pmatrix}, \\ \mathbf{L} &= \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2\sigma} \end{pmatrix}.\end{aligned}\tag{A.4}$$

The transformation starts from the covariance of the sample mean and Bessel-corrected sample variance. Its Jacobian $\mathbf{L}$ leaves the mean unchanged and maps variance to its positive square root.

$$\mathbf{L} \Sigma_{\bar{X},S^2} = \begin{pmatrix} \frac{\sigma^2}{n} & \frac{\mu_3}{n} \\ \frac{\mu_3}{2n\sigma} & \frac{\mu_4 - \frac{n-3}{n-1}\sigma^4}{2n\sigma} \end{pmatrix}.\tag{A.5}$$

Left multiplication by $\mathbf{L}$ divides the second row by twice the population standard deviation.

$$\begin{aligned}\Sigma_{\bar{X},S} &\approx \mathbf{L} \Sigma_{\bar{X},S^2} \mathbf{L}^T, \\ \Sigma_{\bar{X},S} &\approx \begin{pmatrix} \frac{\sigma^2}{n} & \frac{\mu_3}{2n\sigma} \\ \frac{\mu_3}{2n\sigma} & \frac{\mu_4 - \frac{n-3}{n-1}\sigma^4}{4n\sigma^2} \end{pmatrix}.\end{aligned}\tag{A.6}$$

Right multiplication by the transpose of $\mathbf{L}$ scales the second column. Together, the two multiplications divide the mean–variance covariance by twice the population standard deviation and the variance of the sample variance by four times the population variance, yielding the first-order mean–standard deviation covariance matrix.

$$\begin{aligned}A &= \frac{\sigma^2}{n}, \\ B &= \frac{\mu_3}{2n\sigma}, \\ C &= \frac{\mu_4 - \frac{n-3}{n-1}\sigma^4}{4n\sigma^2}.\end{aligned}\tag{A.7}$$

A is the exact variance of the sample mean. B and C are first-order approximations to the mean–standard deviation covariance and the variance of the sample standard deviation, respectively. Their dependence on the third and fourth central moments makes B and C distribution dependent. For a normal population, B is zero; it is generally nonzero for the lognormal and gamma families [3].

The third and fourth central moments reflect distributional asymmetry and the contribution of large deviations from the mean, respectively. The family-specific expressions below are evaluated using the estimated arithmetic means and standard deviations. The reported uncertainty is therefore obtained by substituting parameter estimates into these expressions, even when the underlying variance identities are exact. For the lognormal expressions, $c = 1 + \left(\frac{\sigma}{\mu}\right)^2$.

$$\begin{aligned}
&\text{Normal: } \mu_3 = 0, \quad \mu_4 = 3 \sigma^4; \\
&\text{Lognormal: } \mu_3 = \frac{(c + 2) \sigma^4}{\mu}, \quad \mu_4 = (c^4 + 2c^3 + 3c^2 - 3) \sigma^4; \\
&\text{Gamma: } \mu_3 = \frac{2 \sigma^4}{\mu}, \quad \mu_4 = 3 \sigma^4 + \frac{6 \sigma^6}{\mu^2}.
\end{aligned}$$

$$\begin{aligned}
\mathbf{\Sigma}_D &= \begin{pmatrix} A_D & B_D \\ B_D & C_D \end{pmatrix}, \\
\mathbf{\Sigma}_H &= \begin{pmatrix} A_H & B_H \\ B_H & C_H \end{pmatrix}.
\end{aligned} \tag{A.8}$$

For each population, the sample arithmetic mean precedes the sample standard deviation in the input vector. The corresponding covariance matrices are symmetric.

$$\mathbf{\Sigma}_{\text{par}} = \begin{pmatrix} A_D & B_D & 0 & 0 \\ B_D & C_D & 0 & 0 \\ 0 & 0 & A_H & B_H \\ 0 & 0 & B_H & C_H \end{pmatrix}. \tag{A.9}$$

The matrix $\mathbf{\Sigma}_{\text{par}}$ contains the variances and covariances of the four fitted population summaries. The upper-left and lower-right submatrices correspond to the diseased and nondiseased populations, respectively. Covariances between estimators from different populations are zero under the assumption of independent samples.

$$\mathbf{\Sigma}_x = \begin{pmatrix} A_D & B_D & 0 & 0 & 0 & 0 \\ B_D & C_D & 0 & 0 & 0 & 0 \\ 0 & 0 & A_H & B_H & 0 & 0 \\ 0 & 0 & B_H & C_H & 0 & 0 \\ 0 & 0 & 0 & 0 & u_v^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & u_{\text{meas}}^2 \end{pmatrix}. \tag{A.10}$$

Including prevalence and the evaluation argument alongside the four fitted population summaries extends the input covariance matrix from 4×4 to 6×6 . Their variances occupy matrix elements (5,5) and (6,6), respectively. All additional off-diagonal elements are zero under the assumed independence of the measurement and prevalence uncertainty sources from the population-parameter sampling sources and from each other.

A.5.2. Prevalence uncertainty

Under the assumed cohort sampling model, prevalence and its standard uncertainty are estimated as follows:

$$\begin{aligned}
v &= \frac{n_D}{n_D + n_H}. \\
u_v &= \sqrt{\frac{v(1-v)}{n_D + n_H}}.
\end{aligned}$$

A.5.3. Common measurement uncertainty

The measurement standard uncertainty $u_{\text{meas}}(t)$ is specified by either a linear model, restricted to nonnegative values, or a root-sum-of-squares model. The coefficient b_0 is the constant component and has the measurement unit; b_1 is the dimensionless proportional coefficient. Both coefficients are treated as known. The uncertainty model is evaluated at the diagnostic threshold; u_{meas} denotes the same source uncertainty

where the argument is understood. The assumed joint propagation of measurement uncertainty requires empirical justification; total quality-control variance alone does not establish the covariance structure. Independent analytical variation in individual observations belongs in the measured population distributions or an explicit observation model and must not be counted twice. Gauss–Hermite quadrature separately propagates the measurement component at an estimated optimal threshold.

$$\begin{aligned}
u_{\text{meas}} &= b_0 + b_1 t \\
&\text{or} \\
u_{\text{meas}} &= \sqrt{b_0^2 + (b_1 t)^2}. \\
\mathbf{h}_{\text{meas}} &= \begin{pmatrix} \frac{\partial \text{Se}}{\partial t} \\ \frac{\partial \text{Sp}}{\partial t} \end{pmatrix} = \begin{pmatrix} -f_D(t) \\ f_H(t) \end{pmatrix}.
\end{aligned} \tag{A.11}$$

In equation (A.11) and the subsequent Jacobians, the derivatives of Se and Sp with respect to the numerical evaluation coordinate are evaluated at $\tau = t$. They are respectively nonpositive and nonnegative. The 2×1 sensitivity vector and its transpose determine the measurement covariance contribution without assigning uncertainty to the diagnostic threshold.

$$\begin{aligned}
\boldsymbol{\Sigma}_{\text{Se,Sp;meas}} &= u_{\text{meas}}^2 \mathbf{h}_{\text{meas}} \mathbf{h}_{\text{meas}}^T, \\
\boldsymbol{\Sigma}_{\text{Se,Sp;meas}} &= u_{\text{meas}}^2 \begin{pmatrix} -f_D(t) \\ f_H(t) \end{pmatrix} \begin{pmatrix} -f_D(t) & f_H(t) \end{pmatrix}.
\end{aligned} \tag{A.12}$$

$$\begin{aligned}
\boldsymbol{\Sigma}_{\text{Se,Sp;meas}} &= u_{\text{meas}}^2 \begin{pmatrix} (-f_D(t))(-f_D(t)) & (-f_D(t))f_H(t) \\ f_H(t)(-f_D(t)) & f_H(t)f_H(t) \end{pmatrix}, \\
\boldsymbol{\Sigma}_{\text{Se,Sp;meas}} &= u_{\text{meas}}^2 \begin{pmatrix} f_D(t)^2 & -f_D(t)f_H(t) \\ -f_D(t)f_H(t) & f_H(t)^2 \end{pmatrix}.
\end{aligned} \tag{A.13}$$

The diagonal elements are the squared population densities multiplied by the measurement variance. The off-diagonal elements are the negative product of the two densities multiplied by the same variance. The resulting covariance matrix has rank one whenever that variance is positive and at least one density is nonzero.

The measurement covariance contribution between Se and Sp is nonpositive because their sensitivity coefficients have opposite signs. The source variance and both marginal variance contributions remain nonnegative.

A.6. Calculation of the derivatives

The partial derivatives of each measure with respect to the six inputs are obtained from its derivatives with respect to the three intermediate inputs by the chain rule.

$$\begin{aligned}
 a &= \text{Se}, & b &= \text{Sp}, \\
 J &= a + b - 1, \\
 q &= v a + (1 - v)(1 - b), \\
 p_o &= v a + (1 - v)b, \\
 p_e &= (1 - v) + (2v - 1)q, \\
 \eta &= v + q.
 \end{aligned} \tag{A.14}$$

$$\begin{aligned}
 \text{PPV} &= \frac{v a}{q}, & \text{NPV} &= \frac{(1 - v)b}{(1 - q)}, \\
 \text{FMI} &= a \sqrt{\frac{v}{q}}, \\
 r_v &= \sqrt{v(1 - v)}, & r_q &= \sqrt{q(1 - q)}, \\
 \text{MCC} &= \frac{r_v J}{r_q}.
 \end{aligned} \tag{A.15}$$

The quantities q , p_o , and p_e denote predicted-positive probability, observed agreement, and chance agreement, respectively. The denominator in F_1 is denoted $\eta = v + q$; its square appears in the derivative denominators. For MCC, r_v and r_q denote the square-root factors associated with prevalence and predicted-positive probability, respectively; J retains its definition as Youden's index. PPV, NPV, and FMI retain their established definitions.

$$\begin{aligned}
 q_a &= v, & q_b &= -(1 - v), & q_v &= J; \\
 p_{o_a} &= v, & p_{o_b} &= (1 - v), & p_{o_v} &= a - b; \\
 p_{e_a} &= v(2v - 1), \\
 p_{e_b} &= (1 - v)(1 - 2v), \\
 p_{e_v} &= -1 + 2q + (2v - 1)J.
 \end{aligned} \tag{A.16}$$

The derivatives of the three probabilities follow from their definitions. The derivative of predicted-positive probability with respect to Sp is nonpositive. The prevalence derivative of chance agreement includes both contributions from the product rule.

$$\begin{aligned}
 \mathbf{G} &= \frac{\partial \mathbf{m}}{\partial \mathbf{z}^T} \in \mathbb{R}^{16 \times 3}, \\
 \mathbf{G}_{i \cdot} &= \left(\frac{\partial F_i}{\partial a} \quad \frac{\partial F_i}{\partial b} \quad \frac{\partial F_i}{\partial v} \right) = (G_{ia} \quad G_{ib} \quad G_{iv}).
 \end{aligned} \tag{A.17}$$

TABLE A4. ALL 48 DERIVATIVES WITH RESPECT TO SENSITIVITY, SPECIFICITY, AND PREVALENCE.

Measure	Jacobian element	Expanded scalar derivative
Se	$\frac{\partial \text{Se}}{\partial a}$	1
Se	$\frac{\partial \text{Se}}{\partial b}$	0
Se	$\frac{\partial \text{Se}}{\partial v}$	0
Sp	$\frac{\partial \text{Sp}}{\partial a}$	0

Sp	$\frac{\partial \text{Sp}}{\partial b}$	1
Sp	$\frac{\partial \text{Sp}}{\partial v}$	0
ACC	$\frac{\partial \text{ACC}}{\partial a}$	v
ACC	$\frac{\partial \text{ACC}}{\partial b}$	$(1 - v)$
ACC	$\frac{\partial \text{ACC}}{\partial v}$	$a - b$
PPV	$\frac{\partial \text{PPV}}{\partial a}$	$\frac{v(1 - v)(1 - b)}{q^2}$
PPV	$\frac{\partial \text{PPV}}{\partial b}$	$\frac{v a (1 - v)}{q^2}$
PPV	$\frac{\partial \text{PPV}}{\partial v}$	$\frac{a(1 - b)}{q^2}$
NPV	$\frac{\partial \text{NPV}}{\partial a}$	$\frac{v(1 - v)b}{(1 - q)^2}$
NPV	$\frac{\partial \text{NPV}}{\partial b}$	$\frac{v(1 - v)(1 - a)}{(1 - q)^2}$
NPV	$\frac{\partial \text{NPV}}{\partial v}$	$-\frac{b(1 - a)}{(1 - q)^2}$
DOR	$\frac{\partial \text{DOR}}{\partial a}$	$\frac{b}{(1 - b)(1 - a)^2}$
DOR	$\frac{\partial \text{DOR}}{\partial b}$	$\frac{a}{(1 - a)(1 - b)^2}$
DOR	$\frac{\partial \text{DOR}}{\partial v}$	0
LR⁺	$\frac{\partial \text{LR}^+}{\partial a}$	$\frac{1}{(1 - b)}$
LR⁺	$\frac{\partial \text{LR}^+}{\partial b}$	$\frac{a}{(1 - b)^2}$
LR⁺	$\frac{\partial \text{LR}^+}{\partial v}$	0
LR⁻	$\frac{\partial \text{LR}^-}{\partial a}$	$-\frac{1}{b}$
LR⁻	$\frac{\partial \text{LR}^-}{\partial b}$	$-\frac{(1 - a)}{b^2}$
LR⁻	$\frac{\partial \text{LR}^-}{\partial v}$	0
J	$\frac{\partial J}{\partial a}$	1
J	$\frac{\partial J}{\partial b}$	1
J	$\frac{\partial J}{\partial v}$	0

BA	$\frac{\partial \text{BA}}{\partial a}$	$\frac{1}{2}$
BA	$\frac{\partial \text{BA}}{\partial b}$	$\frac{1}{2}$
BA	$\frac{\partial \text{BA}}{\partial v}$	0
CZ	$\frac{\partial \text{CZ}}{\partial a}$	b
CZ	$\frac{\partial \text{CZ}}{\partial b}$	a
CZ	$\frac{\partial \text{CZ}}{\partial v}$	0
FMI	$\frac{\partial \text{FMI}}{\partial a}$	$\text{FMI} \left(\frac{1}{a} - \frac{v}{2q} \right)$
FMI	$\frac{\partial \text{FMI}}{\partial b}$	$\frac{\text{FMI} (1 - v)}{2q}$
FMI	$\frac{\partial \text{FMI}}{\partial v}$	$\text{FMI} \left(\frac{1}{2v} - \frac{J}{2q} \right)$
κ	$\frac{\partial \kappa}{\partial a}$	$\frac{(1 - p_e) v + (p_o - 1) v (2v - 1)}{(1 - p_e)^2}$
κ	$\frac{\partial \kappa}{\partial b}$	$\frac{(1 - p_e) (1 - v) + (p_o - 1) (1 - v) (1 - 2v)}{(1 - p_e)^2}$
κ	$\frac{\partial \kappa}{\partial v}$	$\frac{(1 - p_e) (a - b) + (p_o - 1) (-1 + 2q + (2v - 1)J)}{(1 - p_e)^2}$
PABAK	$\frac{\partial \text{PABAK}}{\partial a}$	$2v$
PABAK	$\frac{\partial \text{PABAK}}{\partial b}$	$2(1 - v)$
PABAK	$\frac{\partial \text{PABAK}}{\partial v}$	$2(a - b)$
F_1	$\frac{\partial F_1}{\partial a}$	$\frac{2v(v + (1 - v)(1 - b))}{\eta^2}$
F_1	$\frac{\partial F_1}{\partial b}$	$\frac{2va(1 - v)}{\eta^2}$
F_1	$\frac{\partial F_1}{\partial v}$	$\frac{2a(1 - b)}{\eta^2}$
MCC	$\frac{\partial \text{MCC}}{\partial a}$	$\frac{r_v}{r_q} \left(1 - \frac{J(1 - 2q)v}{2q(1 - q)} \right)$
MCC	$\frac{\partial \text{MCC}}{\partial b}$	$\frac{r_v}{r_q} \left(1 + \frac{J(1 - 2q)(1 - v)}{2q(1 - q)} \right)$
MCC	$\frac{\partial \text{MCC}}{\partial v}$	$\text{MCC} \left(\frac{(1 - 2v)}{2v(1 - v)} - \frac{J(1 - 2q)}{2q(1 - q)} \right)$

Rows follow the output order in equation (A.2). The three derivatives for each measure are taken with respect to a , b , and v . Denominators must be nonzero; expressions containing reciprocals or square roots require parameter values in their differentiability domains.

Table A4 supplies all elements of $\mathbf{G}$. The 16×1 output vector is differentiated with respect to the transpose of the 3×1 intermediate vector $\mathbf{z}$, yielding a 16×3 Jacobian.

A.6.1. Chain-rule expansion from three intermediate to six inputs

$$\begin{aligned} a_m &= \frac{\partial a}{\partial m_D}, & a_s &= \frac{\partial a}{\partial s_D}, \\ b_m &= \frac{\partial b}{\partial m_H}, & b_s &= \frac{\partial b}{\partial s_H}. \end{aligned} \tag{A.18}$$

$$\mathbf{K} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}^T} = \begin{pmatrix} a_m & a_s & 0 & 0 & 0 & -f_D(t) \\ 0 & 0 & b_m & b_s & 0 & f_H(t) \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}. \tag{A.19}$$

The first row contains sensitivity coefficients for the diseased summaries and the evaluation argument; the second contains the corresponding coefficients for the nondiseased summaries and the evaluation argument. The third row selects prevalence. Population-family tail formulas supply the mean and standard-deviation derivatives. The threshold coefficients, evaluated at the diagnostic threshold, are minus the diseased density and plus the nondiseased density.

$$\mathbf{M} = \mathbf{G} \mathbf{K} \in \mathbb{R}^{16 \times 6}, \tag{A.20}$$

$$\mathbf{M}_{i.} = (G_{ia} a_m \quad G_{ia} a_s \quad G_{ib} b_m \quad G_{ib} b_s \quad G_{iv} \quad -G_{ia} f_D(t) + G_{ib} f_H(t)).$$

Multiplication of $\mathbf{G}$ by $\mathbf{K}$ applies the chain rule to each measure. The sixth column of $\mathbf{M}$ combines the contributions of Se and Sp to the derivative with respect to the evaluation argument at the diagnostic threshold.

Population-summary derivatives are obtained by differentiating the fitted tail probabilities with respect to the arithmetic mean or standard deviation, including the induced changes in the distribution parameters. For the lognormal family, a change in the arithmetic standard deviation changes both lognormal parameters. Sensitivity coefficients for the evaluation argument are obtained directly from the population densities evaluated at the diagnostic threshold.

A numerical approximation to a smooth scalar derivative is obtained by a central difference, with all other inputs held fixed:

$$\frac{g(x + h \mathbf{e}_j) - g(x - h \mathbf{e}_j)}{2h} \approx \partial g / \partial x_j.$$

The unit vector $\mathbf{e}_j$ identifies the perturbed input, and the positive step h has the same unit as that input. Both perturbed inputs must remain in the admissible domain. Stability under a reduction in step size provides a numerical check, although excessively small steps may amplify rounding effects.

A.7. Uncertainty propagation for a single measure

First-order propagation combines input variances and covariances with the corresponding partial derivatives to obtain the output variance. The combined standard uncertainty is its nonnegative square root. Variance contributions from independent sources are additive; standard uncertainties are not. The scalar expansion below expresses the calculation explicitly.

$$\mathbf{d} = \nabla g = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ d_6 \end{pmatrix}, \quad (A.21)$$

$$\begin{aligned} d_1 &= \frac{\partial F}{\partial m_D}, & d_2 &= \frac{\partial F}{\partial s_D}, \\ d_3 &= \frac{\partial F}{\partial m_H}, & d_4 &= \frac{\partial F}{\partial s_H}, \\ d_5 &= \frac{\partial F}{\partial v}, & d_6 &= \frac{\partial F}{\partial t}. \end{aligned}$$

The gradient contains six partial derivatives. The propagated variance is the quadratic form obtained by multiplying the input covariance matrix by the gradient and premultiplying the result by its transpose.

$$\mathbf{w} = \Sigma_{\mathbf{x}} \mathbf{d} = \begin{pmatrix} A_D d_1 + B_D d_2 \\ B_D d_1 + C_D d_2 \\ A_H d_3 + B_H d_4 \\ B_H d_3 + C_H d_4 \\ u_v^2 d_5 \\ u_{\text{meas}}^2 d_6 \end{pmatrix}. \quad (A.22)$$

Each element of $\mathbf{w}$ is the scalar product of the corresponding row of the input covariance matrix and the gradient. The first element is $A_D d_1 + B_D d_2$, because the remaining elements of that covariance-matrix row are zero. Premultiplication by $\mathbf{d}^T$ gives the output variance.

$$\begin{aligned} u_c(F)^2 &= \mathbf{d}^T \mathbf{w}, \\ u_c(F)^2 &= A_D d_1^2 + 2B_D d_1 d_2 + C_D d_2^2 \\ &+ A_H d_3^2 + 2B_H d_3 d_4 + C_H d_4^2 + u_v^2 d_5^2 + u_{\text{meas}}^2 d_6^2. \end{aligned} \quad (A.23)$$

Symmetry of the population covariance matrices gives two equal covariance terms for each population. Their sums are $2B_D d_1 d_2$ and $2B_H d_3 d_4$ for the diseased and nondiseased populations, respectively. Cross-terms between independent sources are zero.

$$\begin{aligned} V_D &= A_D d_1^2 + 2B_D d_1 d_2 + C_D d_2^2, \\ V_H &= A_H d_3^2 + 2B_H d_3 d_4 + C_H d_4^2, \\ V_v &= u_v^2 d_5^2 = d_5^2 \frac{v(1-v)}{n_D + n_H}, \\ V_{\text{meas}} &= u_{\text{meas}}^2 d_6^2 = d_6^2 [b_0^2 + (b_1 t)^2], \\ u_c(F)^2 &= V_D + V_H + V_v + V_{\text{meas}}. \end{aligned} \quad (A.24)$$

Equation (A.24) decomposes the output variance into source-specific contributions and retains the covariance between the estimated mean and standard deviation within each population. Sampling variance includes the population-parameter contributions and the prevalence contribution. The measurement variance contribution results from propagating the common measurement uncertainty while holding the diagnostic threshold fixed. The combined standard uncertainty is the nonnegative square root of their sum. The measurement-variance expression in equation (A.24) uses the root-sum-of-squares model.

Within-population covariance terms may be positive or negative and are retained in the corresponding variance contributions. Each valid population covariance matrix yields a nonnegative propagated variance, although a covariance term may reduce that variance. Omitting a negative covariance term changes the assumed dependence structure.

For a generic measure F , combined, total sampling, and measurement standard uncertainties are denoted $u_c(F)$, $u_s(F)$, and $u_m(F)$, respectively; the parameter and prevalence sampling components are $u_{\text{par}}(F)$ and $u_{\text{prev}}(F)$. Figure legends use this notation, with the measure argument implicit where omitted.

The combined variance is the sum of the independent source-specific variance contributions, with all within-population covariances retained. Parameter sampling uncertainty comprises the diseased and nondiseased population contributions; total sampling uncertainty also includes the prevalence contribution.

$$u_s(F)^2 = u_{\text{par}}(F)^2 + u_{\text{prev}}(F)^2, \quad u_c(F)^2 = u_s(F)^2 + u_m(F)^2$$

A.8. Joint uncertainty propagation for all sixteen measures

$$\begin{aligned} V_a &= A_D a_m^2 + 2B_D a_m a_s + C_D a_s^2 + f_D(t)^2 u_{\text{meas}}^2, \\ V_b &= A_H b_m^2 + 2B_H b_m b_s + C_H b_s^2 + f_H(t)^2 u_{\text{meas}}^2, \\ C_{ab} &= -f_D(t) f_H(t) u_{\text{meas}}^2. \end{aligned} \tag{A.25}$$

The first three terms of V_a give the contribution from the estimated diseased population mean and standard deviation to the variance of Se; the final term gives the common measurement contribution. V_b contains the corresponding nondiseased population and measurement contributions to the variance of Sp. Under the stated independence assumptions, C_{ab} arises solely from the common measurement uncertainty and is nonpositive because the two threshold derivatives have opposite signs.

$$\begin{aligned} \Sigma_z &= \mathbf{K} \Sigma_x \mathbf{K}^T, \\ \Sigma_z &= \begin{pmatrix} V_a & C_{ab} & 0 \\ C_{ab} & V_b & 0 \\ 0 & 0 & u_v^2 \end{pmatrix}. \end{aligned} \tag{A.26}$$

The intermediate covariance matrix has dimensions 3×3 . Covariances involving prevalence are zero under its assumed independence from the population-parameter sampling and measurement uncertainty sources. Other sampling designs may require additional covariance terms.

$$\begin{aligned} \Sigma_m &= \mathbf{M} \Sigma_x \mathbf{M}^T, \\ \Sigma_m &= \mathbf{G} \Sigma_z \mathbf{G}^T. \end{aligned} \tag{A.27}$$

The two expressions yield the same 16×16 output covariance matrix because $\mathbf{M} = \mathbf{G} \mathbf{K}$. They represent equivalent formulations of uncertainty propagation. The factorization distinguishes the 16×3 Jacobian with respect to intermediate inputs from the 16×6 Jacobian with respect to inputs.

$$\begin{aligned} \text{Cov}(F_i, F_j) &= G_{ia} V_a G_{ja} + G_{ia} C_{ab} G_{jb} + G_{ia} 0 G_{jv}, \\ &\quad + G_{ib} C_{ab} G_{ja} + G_{ib} V_b G_{jb} + G_{ib} 0 G_{jv}, \\ &\quad + G_{iv} 0 G_{ja} + G_{iv} 0 G_{jb} + G_{iv} u_v^2 G_{jv}, \\ \text{Cov}(F_i, F_j) &= G_{ia} V_a G_{ja} + G_{ia} C_{ab} G_{jb} + G_{ib} C_{ab} G_{ja} + G_{ib} V_b G_{jb} + G_{iv} u_v^2 G_{jv}. \end{aligned} \tag{A.28}$$

The expanded product contains nine terms, four of which vanish under the assumed independence of prevalence. Off-diagonal elements are covariances between measures, and diagonal elements are their variances.

$$u_c(F_i)^2 = G_{ia}^2 V_a + 2G_{ia} G_{ib} C_{ab} + G_{ib}^2 V_b + G_{iv}^2 u_v^2. \tag{A.29}$$

The factor of two in equation (A.29) arises from the two equal covariance terms for Se and Sp. When C_{ab} is negative, their combined contribution decreases the propagated variance when the corresponding partial derivatives have the same sign and increases it when they have opposite signs. Omitting this contribution would treat the measurement contributions to Se and Sp as independent and would contradict the stated joint propagation of measurement uncertainty.

$$\begin{aligned} V_{\text{meas}_i} &= u_{\text{meas}}^2 (-G_{ia} f_D(t) + G_{ib} f_H(t))^2, \\ V_{\text{meas}_i} &= u_{\text{meas}}^2 [G_{ia}^2 f_D(t)^2 - 2G_{ia} G_{ib} f_D(t) f_H(t) + G_{ib}^2 f_H(t)^2]. \end{aligned} \tag{A.30}$$

Expansion of the squared derivative with respect to the evaluation argument recovers the common measurement covariance contribution for Se and Sp already included in the propagated variance.

A.8.1. Source-specific expansion in input coordinates

$$\Sigma_{\mathbf{m}} = \mathbf{M}_D \Sigma_D \mathbf{M}_D^T + \mathbf{M}_H \Sigma_H \mathbf{M}_H^T + \mathbf{j}_v u_v^2 \mathbf{j}_v^T + \mathbf{j}_{\text{meas}} u_{\text{meas}}^2 \mathbf{j}_{\text{meas}}^T. \quad (\text{A.31})$$

The diseased and nondiseased Jacobian submatrices comprise the first two and the next two columns of $\mathbf{M}$, respectively. The prevalence and measurement sensitivity vectors $\mathbf{j}_v$ and $\mathbf{j}_{\text{meas}}$ are its fifth and sixth columns. The sixth column contains derivatives with respect to the evaluation argument at the diagnostic threshold. Each source-specific covariance contribution has dimensions 16×16 .

$$\begin{aligned} \Sigma_{\mathbf{m}ij} = & A_D M_{i1} M_{j1} + B_D (M_{i1} M_{j2} + M_{i2} M_{j1}) + C_D M_{i2} M_{j2}, \\ & + A_H M_{i3} M_{j3} + B_H (M_{i3} M_{j4} + M_{i4} M_{j3}) + C_H M_{i4} M_{j4}, \\ & + u_v^2 M_{i5} M_{j5} + u_{\text{meas}}^2 M_{i6} M_{j6}. \end{aligned} \quad (\text{A.32})$$

The expanded expression retains both within-population covariance terms. Equal output indices recover the scalar quadratic form, including the factors of two.

The diagonal elements of the output covariance matrix are the variances of the sixteen measures; the off-diagonal elements are their pairwise covariances. The scalar and joint formulations yield identical variances. Their agreement provides a check of implementation consistency but does not assess the fit of the population model to the observations.

A.9. Conversion of standard uncertainty to an approximate expanded-uncertainty interval

Standard uncertainty is uncertainty expressed as a standard deviation. Expanded uncertainty is obtained by multiplying the combined standard uncertainty by a coverage factor. The coverage factor depends on the specified nominal coverage level and the approximate effective degrees of freedom. Each population's variance contribution retains the covariance between its estimated mean and standard deviation in the calculation of effective degrees of freedom.

The four nonnegative source-specific variance contributions are denoted V_D , V_H , V_v , and V_{meas} , with the covariance between each population's estimated mean and standard deviation retained. Effective degrees of freedom are approximated by [2, 3]:

$$\begin{aligned} V_D &= \nabla g_D^T \Sigma_D \nabla g_D, \\ V_H &= \nabla g_H^T \Sigma_H \nabla g_H, \\ V_v &= g_v^2 u_v^2, \\ V_{\text{meas}} &= g_t^2 u_{\text{meas}}^2. \\ v_{\text{eff}} &= \frac{(V_D + V_H + V_v + V_{\text{meas}})^2}{\frac{V_D^2}{v_D} + \frac{V_H^2}{v_H} + \frac{V_v^2}{v_v} + \frac{V_{\text{meas}}^2}{v_{\text{meas}}}}. \end{aligned}$$

The sampling degrees of freedom are $v_D = n_D - 1$, $v_H = n_H - 1$, and $v_v = n_D + n_H - 1$. Assigning $v_{\text{meas}} = \infty$ treats the common measurement standard uncertainty as known without appreciable uncertainty. This assumption does not follow solely from classification as a Type B evaluation. For a nominal coverage level of $1 - \alpha$:

$$k = t_{v_{\text{eff}}, 1-\alpha/2}, \quad U = k u_c(F)$$

Here k is the $1 - \alpha/2$ quantile of Student's t distribution with v_{eff} degrees of freedom, and U is the expanded uncertainty. If an uncertainty component has infinite degrees of freedom, its term in the denominator of the effective-degrees-of-freedom expression is zero, but its contribution to the combined variance remains.

The two-sided expanded-uncertainty interval is:

$$I_{1-\alpha}(F) = [F - U, F + U]$$

At specified diagnostic thresholds, these pointwise expanded-uncertainty intervals are based on a first-order Wald approximation. At an estimated optimal threshold, the common measurement uncertainty component is propagated nonlinearly by Gauss–Hermite quadrature, conditional on that fixed selected value. The notebook restricts the interval endpoints to each measure's admissible range. Restriction to the admissible range does not ensure nominal coverage or remove the limitations of the approximation. Standard uncertainties are unaffected by this restriction.

A.10. Evaluation of uncertainty at an estimated optimal threshold

In *DiagAccU.nb*, Wolfram NMaximize maximizes the selected measure's point estimate subject to $t > m_H$, without a prescribed upper threshold bound. The fitted population distributions remain fixed. Conditional on the estimated optimal threshold, first-order sampling variance is combined with the common measurement variance obtained by nine-point Gauss–Hermite quadrature using a normal integration rule centered at the fixed estimated optimal threshold.

The resulting interval quantifies uncertainty in the diagnostic accuracy measure conditional on the estimated optimal threshold subject to $t > m_H$. That selected value is held fixed in the uncertainty calculation. Optimal threshold uncertainty, arising from estimating and selecting that value, is excluded.

At a smooth interior maximum, the first derivative with respect to the deterministic threshold is zero. First-order propagation may therefore yield a zero measurement contribution while omitting higher-order contributions. Nonlinear propagation estimates the measurement variance from weighted evaluations of the complete measure at quadrature nodes.

The quadrature calculation uses a normal integration rule centered at the fixed estimated optimal threshold, with scale given by the measurement standard uncertainty. Nine-point Gauss–Hermite quadrature approximates expectations by nine weighted function evaluations. For nodes and weights defined with respect to the weight $\exp(-x^2)$, the evaluation nodes and normalized weights are:

$$\tau_i = t^* + \sqrt{2} u_{\text{meas}}(t^*) x_i, \quad p_i = w_i / \sqrt{\pi}, \quad i = 1, \dots, 9.$$

The complete diagnostic accuracy measure is evaluated at each numerical node, with the fitted population summaries and prevalence held fixed. The measurement standard uncertainty is evaluated at the estimated optimal threshold and held fixed throughout the integration. The weighted variance of the nine evaluations gives the measurement variance contribution. Its sum with the first-order sampling variance gives the combined variance, whose nonnegative square root is the combined standard uncertainty.

The estimated optimal threshold remains fixed at the center of the conditional analysis. The weighted mean used to calculate the measurement variance may differ from the measure evaluated at that threshold. Population distributions are not re-estimated, and threshold selection is not repeated. Optimal threshold uncertainty can be assessed by repeating these steps within an appropriately specified bootstrap. For positive-support families, the evaluation arguments follow the support conventions of the fitted distributions.

A.11. Implementation and reproducibility

For first-order scalar propagation, a single AroundReplace operation propagates the final scalar measure from a VectorAround object for the four correlated population summaries, an optional Around input for prevalence, and one Around input for the evaluation argument [4, 5]. Its nominal value equals the diagnostic threshold; the uncertainty assigned to this computational input represents the common measurement uncertainty. DiagnosticAccuracyAllComponents, DiagnosticAccuracyAllConfidenceIntervals, and DiagnosticAccuracyJointResult provide reproducible component, interval, and covariance records, respectively.

DiagAccU.nb (Supplemental file 2) is the executable computational source. *DiagAccUCalculations.nb* (Supplemental file 3) provides display-based mathematical derivations and does not contain an independently executable validation suite. *DiagAccUInterface.dpdf* (Supplemental file 4) describes the controls and view-specific options. Sections A.5–A.8 expand the covariance products and all 48 derivatives with respect to the intermediate inputs.

For first-order propagation at a specified threshold, implementation checks assess agreement between individual variances and the diagonal of the joint covariance matrix, addition of covariance matrices for individual uncertainty sources, symmetry, and nonnegative source-specific variances. An independent

lognormal calculation using the formulas specified here and numerical derivatives reproduced the principal fixed-threshold results. These checks address implementation consistency under the stated model. Interval coverage, model fit, transportability, and optimal threshold uncertainty require separate assessment.

A.11.1. Controlled covariance comparisons

For the population-summary comparison, retain all marginal variances and set only the two within-population mean–standard deviation covariances to zero before propagating the complete input covariance. For the measurement covariance comparison, first form the sensitivity–specificity covariance contribution induced by common measurement uncertainty, then set its off-diagonal entries to zero while retaining its marginal variances. Propagate that modified matrix through the diagnostic-measure derivatives. The first comparison changes combined uncertainty; the second isolates the measurement contribution. Both retain the input estimates and resulting point estimates.

These comparisons isolate the consequences of particular dependence assumptions; they do not establish the empirical covariance structure. They are not Monte Carlo experiments or repeated-sampling coverage studies. Numerical results are reported in the main manuscript rather than this mathematical appendix.

A.11.2. Relationship to the earlier software

Bayesian Diagnostic Insights and *DiagAccU* are distinct implementations. The present specification concerns covariance among population summaries and the joint vector of threshold-dependent measures. Shared distribution families and the law of uncertainty propagation are inherited methodological components. Differentiation with respect to a common evaluation argument propagates measurement uncertainty through both distribution tails; *DiagAccU* additionally represents this dependence in the source-specific and joint covariance matrices.

Appendix B. Abbreviations, notation, and software

B.1. List of abbreviations

DAM: diagnostic accuracy measure

CI: confidence interval; the software label for an approximate pointwise expanded-uncertainty interval

GUM: Guide to the Expression of Uncertainty in Measurement

OGTT: oral glucose tolerance test

FPG: fasting plasma glucose

ROC: receiver operating characteristic

B.2. Notation

B.2.1. Populations

H: nondiseased population

D: diseased population

B.2.2. Test outcomes

TN: true negative

TP: true positive

FN: false negative

FP: false positive

B.2.3. DAMs

Se: sensitivity

Sp: specificity

PPV: positive predictive value

NPV: negative predictive value

ACC: overall diagnostic accuracy

DOR: diagnostic odds ratio

LR^+ : positive likelihood ratio

LR^- : negative likelihood ratio

J : Youden's index

BA: balanced accuracy

CZ: concordance probability

FMI: Fowlkes–Mallows index

κ : Cohen's kappa

PABAK: prevalence- and bias-adjusted kappa

F_1 : F_1 score

MCC: Matthews correlation coefficient

B.2.4. Parameters

μ_P, m_D, m_H : population arithmetic mean and supplied sample means, respectively; P denotes D or H

σ_P, s_D, s_H : population standard deviation and supplied Bessel-corrected sample standard deviations, respectively

n_D, n_H : sample sizes of the diseased and nondiseased populations, respectively

v : prevalence

t : specified fixed diagnostic threshold; τ : numerical CDF evaluation coordinate with nominal value $\tau = t$

t^* : estimated optimal threshold subject to $t > m_H$, held fixed in the conditional uncertainty calculation, which excludes optimal threshold uncertainty

$1 - \alpha$: nominal confidence level

B.2.5. Functions and relations

$u_s(F)$: total sampling standard uncertainty of measure F

$u_m(F)$: measurement standard uncertainty of F obtained by propagation; $u_{\text{meas}}(t)$: measurement input standard uncertainty evaluated at the diagnostic threshold

$u_c(F)$: combined standard uncertainty of F

$V_D, V_H, V_v, V_{\text{meas}}$: variance contributions from independent uncertainty sources

$f_D(t), f_H(t)$: population-specific probability densities evaluated at the diagnostic threshold

$F_D(t), F_H(t)$: population-specific cumulative distribution functions evaluated at the diagnostic threshold

q : predicted-positive probability; p_o : observed agreement; p_e : chance agreement

Σ_x : covariance matrix of the six inputs; J : output Jacobian

$I_{1-\alpha}(F)$: approximate expanded-uncertainty interval for F

$\text{Var}(F)$: variance of F ; $\text{Cov}(A, B)$: covariance of A and B

ν_{eff} : effective degrees of freedom calculated from source-specific variance contributions; k : coverage factor;

$U = k u_c$: expanded uncertainty

B.3. Input

B.3.1. Range of input parameters

t : $\max(0.01, \min(m_H - 6s_H, m_D - 6s_D)) - \max(m_H + 6s_H, m_D + 6s_D)$

n_D : 2 – 10,000

m_D : 0.1 – 10,000

s_D : 0.01 – 1,000

$n_{\bar{H}}$: 2 – 10,000

$m_{\bar{H}}$: 0.1 – 10,000

$s_{\bar{H}}$: 0.01 – 1,000

v : 0.001 – 0.999

n_U : 20–10,000.

b_0 : $0 \leq b_0 \leq s_H$

b_1 : 0–0.1

$1 - \alpha$: 0.900–0.999

The diagnostic threshold t , both population means m_D and m_H , both standard deviations s_D and s_H , and b_0 must use the same measurement unit; b_1 is dimensionless. The measurement standard uncertainty and the numerical evaluation coordinate use that same unit.

n : 20–20,000 in views with a total-sample-size control. In the sample-size profile, the total sample size is varied automatically while prevalence is held fixed at the selected value.

B.3.2. Additional input options

B.3.2.1. Plots

Extended and limited settings change the range of the horizontal axis. Their limits depend on the selected view; they do not define the optimization domain or establish interval coverage.

B.3.2.2. Tables

Tables provide a decimal-display control from 1 to 10, with a default of 3. This changes displayed precision, not the precision of the underlying calculation.

B.3.3. Program controls

The program uses a tabbed interface for inputs, estimates, uncertainty components, and profiles. Each terminal view retains its own controls and defaults; settings are not synchronized globally. Analyses should record the values used in each view.

Input values can be selected with menus or adjusted with sliders. Modifier keys allow finer slider adjustments; the available controls are described in the interface documentation.

Plot interaction depends on the Wolfram Front End and the selected view. The interface documentation describes the available controls and ranges of the horizontal axes.

B.4. Software availability and requirements

Software name: *DiagAccU*

Version: 2.0.1

DOI: [10.5281/zenodo.22843673](https://doi.org/10.5281/zenodo.22843673)

Project home page: <https://www.hcsl.com/Tools/DiagnosticAccuracy/>

Computational source notebook: *DiagAccU.nb*. Available at:
<https://www.hcsl.com/Tools/DiagnosticAccuracy/DiagAccU.nb>

Programming language: Wolfram Language

Operating systems: Microsoft Windows 10+, Linux 4.18+, Apple macOS 13+

Hardware requirements: x86-64-v2 and higher or Apple Silicon microarchitecture CPU, and 16 GB+ RAM, recommended.

Software citation: Chatzimichail RA, Chatzimichail T, Hatjimihail AT. *DiagAccU*: a unified framework for uncertainty estimation of diagnostic accuracy measures. Version 2.0.1. Hellenic Complex Systems Laboratory; 2026. DOI: <https://doi.org/10.5281/zenodo.22843673>.

License: [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](https://creativecommons.org/licenses/by-nc-sa/4.0/). The supplied source and reference implementation are intended for inspection and research use.

B.5. Reproducible use and file roles

DiagAccU.nb serves as the computational source and should be evaluated in a fresh Wolfram session before numerical results are requested. Reproducible computation requires documentation of the population distributions, arithmetic summaries, sample sizes, prevalence interpretation, diagnostic threshold, common measurement-uncertainty model, coefficient values, source degrees of freedom, and interval level. Record the empirical basis for the assumed measurement covariance structure and how independent analytical variation is represented without double counting. Concentrations, the threshold, and the measurement standard uncertainty must use consistent units. Comparisons across output views or covariance specifications should use identical recorded input settings.

The source provides functions for calculating scalar uncertainty components, approximate intervals, and the joint output covariance matrix. The manuscript tables specify the illustrative settings, the companion notebook presents the derivations, and the interface guide describes the controls. Reproduction of

calculations from the specified settings does not constitute reconstruction of the participant selection, distribution fitting, or analytical adjustment procedures used to obtain the illustrative summaries.

First Published: September 21, 2025

Revised: September 25, 2026